

Asymmetric information and trading strategies: testing behavior on the primary and secondary T.-Bond markets around auction days.

F. Drudi and M. Massa
ECB*and Insead [†]

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[†]Corresponding author: Massimo Massa, Department of Finance, INSEAD, Boulevard de Constance, Fontainebleau, 77305, Cedex, France. Tel: 33-1-60724481. Fax: 33-1-60724045. E-mail: massimo.massa@insead.edu.

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Abstract

We study how dealers behave when there exist parallel markets for the same asset that are characterized by very different degrees of transparency. We show that the optimal trading strategy, depending on the features of the less transparent market, may involve price manipulation. Informed dealers may refrain from trading in the more transparent market in order to exploit their informational advantage in the less transparent one, or they may use the more transparent market in order to manipulate prices.

We prove that both strategies (hiding and price manipulation) increase market depth in the more transparent market.

We empirically test these conclusions on the Italian Treasury bond market. We estimate dealers strategies jointly on both the primary and the secondary markets. We show that informed dealers place sell orders with other dealers at the time they have higher informational advantage. At the same time they aggressively place bids in the primary market and buy back when the primary market closes. This strategy generates losses in the more transparent market (secondary market) for the period when the less transparent market (primary market) is open and then produces gains once the possibility of affecting the primary market is over.

We show that market depth in the more transparent market increases as a results of these strategies.

Keywords: Parallel markets, strategic behavior, empirical microstructure, interdealer-trading, information.

JEL classification: G11, G12, G14, G18.

1 Introduction

How do traders behave when faced with the possibility of trading in two different markets characterized by different degrees of transparency? How would the simultaneous existence of a less transparent market affect the market depth of the more transparent one?

The first question deals with the problem of identifying the optimal trading strategy of a dealer who is endowed with some short lived information and can choose between trading in a very transparent market where his trade will reveal his information and a less transparent market, where his positions are not revealed.

The dealer may immediately exploits his informational advantage by aggressively operating in both markets. In doing so he would reveal part of his information to the other dealers and influence their behavior in the more transparent market as well as in the less transparent one. Alternatively, he could refrain from trading in the more transparent market in order to exploit his informational advantage in the less transparent one. In doing so, he would miss the opportunities provided by the possibility of immediately exploiting his information in both markets and would effectively place himself out of one market (the more transparent one). However, he would be able to better exploit his informational advantage in the less transparent market.

Finally, the informed dealer may use the more transparent market in order to manipulate prices. That is, he can take advantage of the fact that the other dealers react to his trading in the more transparent market in order to shape their expectations about the value of the asset about which he has superior information. If he knows that the value is higher than the one currently impounded in the price and it is due to rise, he may want to short sell the asset in the more transparent market. In this way he would further reduce competitors' expectations. The dealer would, at the same time, aggressively stock himself up in the less transparent market where his transaction would go largely unnoticed. In this case the trade-off would be between the short terms losses accrued in the more transparent market in order to affect competitors' expectations and the longer term gains arising out of the purchase of the same asset in the less transparent market at very convenient below-market prices.

The choice between these three types of strategies will depend on the characteristics of the market, the ability of the informed trader to affect market expectations

through his trades in the more transparent market, the impact of his trades in the less transparent market and the informational uncertainty of the other competing traders.

The type of strategy chosen by the informed dealer will have relevant implications in terms of market depth. It could dry up liquidity, by inducing the dealers to concentrate their transactions in the less transparent market, and reduce market depth. Alternatively, it could spur additional trade (e.g., manipulative trade) which would raise liquidity and increase market depth.

We study this problem by modelling the strategic behavior of informed dealers in the presence of parallel markets *on the same asset, but with very different degree of transparency*. In particular, we focus on dealers' behavior in the secondary market of T.-Bonds at the times when the primary market (i.e., auction of T.-Bonds) is open.

The auction of Treasury bonds is an opportunity for the informed dealers to trade on the basis of their superior information without a direct and immediate impact on prices. It is also an event that happens with regular and publicly known timing. This provides an experiment that does not have any equivalent in the equity market and which offers a unique opportunity to study dealers' strategies. The auction not only generates information, but also provides the most informed dealers with a way to operate without revealing their information to the other market participants.

We use a unique dataset on the Italian bond market which, at a disaggregated level, records the behavior of *each single* Treasury bond dealer in *both* the primary and the secondary markets. This allows us to identify the type of strategy informed traders adopt in order to exploit their informational advantage.

1.1 Relation with the literature

Out of the extensive literature on strategic trading (Kyle 1985, Holden and Subrahmanyam 1992, Back 1992, Foster and Viswanathan 1993, Back, Cao and Willard 1997, Vayanos 1999), a strand has burgeoned that has focused on multi-market trading. Chowdry and Nanda (1991) study differences in market liquidity and show that, when a security trades at multiple locations simultaneously, the greater the proportion of liquidity trading by large traders who can split their orders across markets, the larger is the correlation between volumes in different markets and the smaller is the informativeness of prices. One market emerges as the dominant loca-

tion. Lyons (1996), Porter and Weaver (1996) and Naik et al. (1997) analyze how transparency affects the behavior of the market players.

Biais (1993), compares centralized and fragmented markets, such as floor and telephone markets and analyzes differences in bidding strategies reflecting differences in market structures. While the equilibrium number of dealers is shown to be increasing in the frequency of trades and the volatility of the value of the asset, the spread is more volatile in centralized than in fragmented markets. Madhavan (1995) shows that large traders who place multiple trades can benefit from the absence of trade disclosure in a fragmented market, as can dealers who face less price competition than in a unified market. Madhavan (1996) shows that market transparency can actually increase price volatility and lower market liquidity.

Pagano and Roell (1996) investigate whether greater transparency enhances market liquidity by reducing the opportunities for taking advantage of uninformed participants. By comparing the price formation process in several stylized trading systems with different degrees of transparency, they find that greater transparency generates lower trading costs for uninformed traders on average, although not necessarily for every size of trade.

More recently, Bloomfield and O'Hara (1999), by using laboratory experiments, show that trade disclosure increases the informational efficiency of transaction prices, but also increases opening bid-ask spread. Trade disclosure benefits market makers at the expense of liquidity traders and informed traders. Also, Bloomfield and O'Hara (2000) show that, while more transparent markets suffer from the competition with less transparent ones, the informational advantage of low-transparency dealers decreases as their number increases while the high-transparency dealers get increasing benefits. Therefore, the existence of a low-transparency trading mechanism does not necessarily extinguishes a high-transparency one.

We contribute to this literature by directly inspecting the mechanism through which the existence of a less transparent market affects strategic dealers' behavior in the more transparent one and analyzing how this affects the market depth of the more transparent market. In particular, we prove that, depending on the degree of informational uncertainty in the more transparent market and on the market impact of the informed dealers' trade in the less transparent one, informed dealers may either hide their informational advantage or resort to price manipulation. Moreover, we show that market manipulation may actually increase market depth in the more transparent market.

We consider an informational event that not only generates information, but also provides an alternative market where investors can operate without revealing their information to the other market participants: the Treasury auctions. We will not directly study the auction in itself,¹ but will instead focus on dealers' joint behavior in the two parallel markets. Our approach relates to the two existing models of strategic trading around an auction: the brought forward by Chatterjea and Jarrow (1998) and the one of Visvanathan and Wang (2000). The former focuses on manipulative behavior and considers how traders use their knowledge of the order flows to manipulate the market, corner the auction and squeeze the short positions. Visvanathan and Wang show how dealers' optimal inventory policy is affected by the auction and do not explicitly deal with its informational dimension. Indeed, they only consider the effects of the interdealer trading after the auction.

Unlike Visvanathan and Wang, we will explicitly focus on the relation between informed trading before and after the auction and on whether it modifies investors' behavior impacts on the secondary market. Also, unlike Chatterjea and Jarrow, we will focus only on the case where the same bond being auctioning off is already traded in the secondary market in such a large volume that there is no possibility of squeezes.² Controlling for squeeze allows us to focus on asymmetric information issues.

In the empirical section we confirm the theoretical intuitions of the model, by finding evidence of price manipulation in the secondary market when the primary market is open. Furthermore, we find that such manipulation is consistent with greater market depth. In particular, we show that informed dealers resort to the interdealer market to place sell orders at the time they have higher informational advantage. At the same time they aggressively place bids in the primary market and buy back when the primary market closes. This strategy generates losses in the more transparent market (secondary market) for the period when the less transparent market (primary market) is open and then produces gains once the possibility of affecting the primary market is over.

To our knowledge this is also the first empirical analysis of dealers's simulta-

¹There is already a very big literature that deals with it (Cammack, 1991, Bikchandani and Huang 1993, Back and Zender 1993, Bikchandani, Edsparr and Huang, 1993, Simon, 1994, Nyborg and Sundaresan, 1996, Wang and Zender 1999).

²Squeeze originates in imperfectly competitive markets where intermediaries who buy bonds for themselves (e.g., to cover short positions) or on behalf of their customers, and who cannot do so at the auction, are forced to obtain them on the secondary market at higher prices (Jegadeesh, 1993).

neous operating in two different markets characterized by strikingly different levels of transparency, but trading the same asset. Furthermore, being such an asset a T.-Bond, this is also the first paper that studies bond trading behavior on both the primary and secondary market, with data disaggregated at dealer's level. Indeed, few papers have directly analyzed such a market, due to the lack of good quality data (Balduzzi, Elton and Green, 1997, and Fleming and Remolona, 1999). We have, instead, data disaggregated at the individual level for both the primary and secondary market.

The paper is articulated as follows. In Section 2 we describe the institutional features of the market under consideration. In Section 3 we model dealers' strategic behavior and lay-out the empirical restrictions. In Section 4 we describe the datasets and the variables used. In Section 5 and 6 we empirically test the restrictions and provide evidence of strategic behavior and of its impact on market depth. A brief conclusion will follow.

2 A description of the parallel markets

We consider the Italian Treasury bond market. The fact that the Italian market is a centralized and regulated one, where the transactions are concentrated, allows us to observe the behavior of the market as a whole. This stands in stark contrast to the other Treasury bond markets, including the US, where the transactions are mostly over-the-counter, and therefore any analysis would have to be limited to a subsection of the market.

An additional advantage is provided by the fact that in the Italian primary market most of the auctions are "re-openings", that is, the same type of bond is issued repeatedly. This important feature, absent in the US data, can be exploited to test the asymmetric behavior of the informed agents in the market, controlling for the possibility of "squeeze", as the same bond is already traded in the secondary market.

- *The secondary market* (Mercato Telematico dei titoli di Stato, M.T.S.) is a screen-based market. Transactions take place between 9.00 a.m. and 5.00 p.m. Three types of dealers are allowed to operate there: "ordinary dealers", "ordinary market makers" and "specialists". There is an average of 360 ordinary dealers, representing banks, investment firms and insurance companies. Out of these dealers, a small group (around 40 on average) are market makers, that is, they commit them-

selves to trade continuously on the market,³ posting a bid and an ask. Ordinary dealers can place orders only with the market makers and cannot post bid and ask prices. Some market makers, defined as "primary market makers" or "specialists", are subject to more stringent requirements. These intermediaries commit themselves to trade a minimum percentage of each type of bond⁴ and to participate in the auctions, purchasing a minimum amount (percentage in terms of total number auctioned off for each type of bond). In the period covered by this study (1994-1996), they were allowed to borrow at a particularly convenient rate at the discount window of the Bank of Italy and enjoy other additional benefits. Since now on we will use the term "ordinary market maker" to define the market makers who are not specialists and the term "dealer" as a generic term to define a market participant.

Each trader (ordinary dealer, ordinary market maker and specialist) has access to a screen where he can observe the bid and ask prices the market makers post, and the maximum number of bonds they commit themselves to trade. Market makers are not anonymous. That is, the name of the market maker appears on the screen next to the bid and ask prices he is posting. Each market maker knows the identity only of the counterpart he is trading with.

The transactions are executed at the posted price. When a transaction takes place, the name of the market maker "hit" blinks, signaling to the market that he is trading and the price at which this transaction takes place. The volume of the transaction is never revealed, except in the rare case where a market maker receives an order equal to the maximum number he has committed himself to trade. In this case, he automatically withdraws from the market for a period not exceeding 60 seconds. This withdrawal is the only signal the market receives about the size of the transaction.

Analogously to the FX market described by Lyons (1995), the slow diffusion of information via interdealer trade is facilitated by the absence of trade reporting. Only aggregate figures for the whole market are available at the end of the day. The screen-based system is transparent to the general public, and the best bid and ask prices are reported also on a specific page by Reuters.

All the transactions are settled through a settlement system owned and operated by a company that acts as a subsidiary of the Central Bank (SIA). The transactions are also continuously monitored by the Central Bank itself which has to check if

³They can withdraw temporarily from the market if they receive an order equal to the maximum number of bonds they are committed to trade ("depth").

⁴1% of the total number of transactions in most liquid bonds and 3% of the less liquid ones.

the market makers meet the requirements in terms of continuous posting of bid-ask prices, the minimum number of transactions executed per category of bond and the size of the bid-ask spread. The tick-size in the market is 1 basis point.⁵

Medium- and long-term bonds are traded on the secondary market and comprise mainly: T. Notes, T. Bonds and financially indexed bonds (F.I. Bonds). In particular, T. Bonds (Buoni del tesoro Poliennali or BTP) are medium- and long-term coupon bonds. Financially indexed bonds (Certificati di Credito del tesoro or CCT) are medium-long term coupon bonds with the value of the coupon indexed to short-term T. Bills. T. Notes (Certificati del Tesoro a Zero Coupon or CTZ) are 2-year zero-coupon bonds. In Table 1, we report descriptive statistics on the secondary market. They include: the number of transactions and trading volume⁶ broken down according to the type of bond and the type of dealer who originated them.

- The *primary Treasury bond market* is based on uniform type auctions. The schedule of the auctions is communicated at the beginning of the year and then the Treasury announces the number of bonds that will be auctioned off one week prior to each auction. Since February 1994, the intermediaries participating in the auction can submit their demand schedules through a computerized telematic system (Rete Nazionale Interbancaria, R.N.I.) before 1.00 p.m. The official auction results are communicated to the market in the afternoon and include the quantities bid, the number allocated and the allotment price. The time interval between 1.00 p.m.⁷ (deadline for the submission of the demand schedules) and the announcement of the auction results is variable and has decreased over time, progressively shrinking from 4 hours at the beginning of 1994 to 30 minutes during the most recent auctions in the sample. This reduction in time can be attributed to the increased ability of the Treasury to aggregate demand schedule and to timely communicate the results of the auction. The adoption of the new system has enabled the intermediaries to react more quickly to new market information, allowing them to formulate their demand schedules on the basis of information available just before the deadline. There are no restrictions analogous to the "35% rule" existing in the US that limits the volume

⁵For example the price can change from 103.42 to 103.43. It is interesting to note that the average underpricing at the auction is around 3.7 basis points.

⁶Volume is defined as the number of transactions multiplied by their nominal values expressed in liras.

⁷Previously, their demand schedules were hand delivered to the Bank of Italy. The new screen-based telematic system was adopted in February 1994 and became compulsory for all the intermediaries in October 1994.

that a single dealer can purchase at the auction.

To increase the depth of the market, the Treasury organizes re-openings where the same type of bond is issued repeatedly. That is, the new batch of the bond being auctioned off is harmonized with the previously issued batches of the same bond traded in the secondary market. Accrued interest, residual life and tax treatment are designed in a way that the new batch of the bond is perfectly identical in value to the ones already issued. Given that bonds are traded on the secondary market immediately following the first issue, there is no need of a when-issued market.⁸ Re-openings represent more than 85% of the total auctions. Given that the Treasury announces the time schedule of the auctions at the beginning of the year, market participants know in advance if and when the bond on auction will be re-opened. The time between two re-openings varies, depending on the type of bond and on the period during the sample. At the beginning of the sample period, the time between two re-openings is, on average, equal to a month and decreases over time. Toward the end of the sample, the interval between two re-openings is, on average, equal to two weeks. A repo market based on over-the-counter transactions exists, but is not very deep.

The existence of the re-openings makes it hard for any intermediary to "corner the market" by squeezing out the other intermediaries. Indeed, to do this, not only should he be allotted most of the bonds auctioned off, but he should also purchase most of those already outstanding in the secondary market.

The auction day is organized as follows: there are two relevant moments: 1.00 p.m. and the time the Treasury announces the results of the auction ("result announcement time"). Trading time in the secondary market can, therefore, be divided into three periods: the interval 9.00 a.m.-1.00 p.m., which corresponds to the time during which demand schedules are submitted and before the official auction deadline of 1.00 p.m. (henceforth the *before* period), the period between 1.00 p.m. and the moment the Treasury announces the results⁹ of the auction (henceforth the *during* period), and the interval between the announcement time and 5.00 p.m., the end of the trading day (henceforth the *after* period). While the *before* period is fixed, the *during* period and the *after* period are variable, with a length depending

⁸Indeed, a not very deep or active "gray market" exists, but only for the issue of the first batch of the new bond, and data are not of the same quality as those used in this paper.

⁹The Treasury announces the auction price, the total volume demanded and the total volume allotted. Each single participant knows the number of bonds allotted to him, but is not informed as to the number demanded by and allotted to the other participants at the auction.

on the time the Treasury announces the results. The Bank of Italy enacts the physical communication of the results to the auction participants via the R.N.I. network system and to the market by inputting the results into the Bank of Italy's pages on Reuter. Each single participant in the auction is informed of the number of bonds he has been allotted, while the volume assigned is revealed to the market as a whole. The auction price and the cover ratio, that is, the ratio between the number of bonds demanded and the number allotted at the clearing price, are also publicly announced.

All the dealers are allowed to participate in the auction, but in general most of the bidding is done by the specialists and the ordinary market makers. In Table 2, Panel A we report the "degree of participation" of each institutional category (specialists, ordinary market makers and ordinary dealers). In particular, the table reports the ratio between the sum of the submitted (winning) bids of the groupings made of the first 10, 15 and 20 biggest dealers belonging to that category and the total market submitted (winning) bids.

Table 2, Panel B, reports some statistics on the concentration relative to all the re-opening auctions of medium- and long-term T. Bonds placed through the screen-based telematic auction mechanism. We use the Herfindhal and Entropy indexes and measures of concentration based on the share of total bids (submitted or winning) of the biggest dealer over total (C10 and C20). The numbers clearly show a strong degree of concentration among the biggest dealers (defined, within each category, in terms of quantities allotted at the auction).

3 The model

We consider the case where the less transparent market is open and compare it to the case in which it is closed. We assume that there is only one asset (a Treasury bond). It is simultaneously traded in the more transparent market (secondary market) and auctioned off in the less transparent market (primary market).

Assumption 1 The stages of the game.

There are two periods, t_1 and t_2 . These correspond to the interval before the auction deadline (morning) and the interval after the auction deadline (afternoon). In the first period both the primary and the secondary markets are open. In other words, dealers simultaneously trade in the secondary market and submit their bids

at the auction. Given that the auction is a sealed bid, information will not be immediately revealed to the market. The auction deadline coincides with the end of this first period.

During the second period, the primary market is closed (i.e., the Treasury screens the bids at the auction) and dealers keep on trading in the secondary market. No public information is yet available about the results of the auction. Finally, the Treasury announces the results of the auction. At the time of the announcement full revelation takes place and the price in the secondary market adjusts to the terminal value of the bond (V). This is the value the bond will have at the end of the auction day when the market has fully incorporated into the prices customers' demand at the auction.

Assumption 2 Type of dealers.

There are four classes of dealers: liquidity traders, uninformed investors, informed dealers and market makers.

The *liquidity traders* operate in the secondary market for liquidity purposes and to rebalance their portfolios. We define their demand as: u_t . We assume, with no loss of generality, that in each period liquidity traders' demand in the secondary market is normally distributed with zero mean and variance equal to Σ_u . That is, $u_t \sim N(0, \Sigma_u)$, for $t = 1, 2$.¹⁰

Liquidity traders also participate to the auction by submitting bids. Their demand schedule is not influenced by informational motives and it is simply a negative function of the prices at the auction. That is, they submit conditional demand schedules. We can therefore represent their demand as:

$$X_a^l = -\beta^l P_a, \quad (1)$$

where β^l is the sensitivity of their demand to the auction price (P_a). The liquidity traders can be thought of as simply intermediating the demand at the auction of the terminal investors. They post bids at the auction at the prices they have been asked to by the terminal investors.

The *uninformed investors* are risk neutral dealers who try to exploit "demand pressure" at the auction. They do not have full information about the price at the

¹⁰There is no reason to assume that liquidity trading differs on average in the two periods or that its variance changes. If the variance were different in the two periods and informed investors knew it, their trading strategies would impound such information (see Spiegel and Subrahmanyan, 1993).

auction, but they try to infer it by observing the prices in the secondary market and the orders they receive by the informed dealers they trade with before the auction. Their speculative demand (X_a^u) is therefore a function of both the prices in the secondary market in the morning and the price at the auction.

That is, they observe prices in the secondary market in the morning and try to infer what the value of the asset will be. Then, they submit a demand schedule that is a function of the difference between the actual price in the market and the expected value of the asset they infer by observing prices in the secondary market in the morning. In particular, they maximize:

$$Max_{x_1} \{E^u [V - P_a] X_a^u | P_1\} \quad (2)$$

$$s.t. \quad : \quad P_a = X_a^i + X_a^u + X_a^l - 1. \quad (3)$$

That is, they account for the impact of their bids at the auction.

The *informed dealers* have an informational advantage about what the value of the bond will be at the end of the day (V). For simplicity we assume that they know the terminal value of the bond from the outset. The informed dealers are risk neutral and must decide how best to exploit their informational advantage by splitting their trade between the primary and the secondary market, before and after the auction. They have to trade-off the pros and cons of trading in two markets characterized by a different degree of transparency. Informed dealers' bids in the primary market (X_a^i) and demand in the secondary market in the two periods (X_1^i and X_2^i) are therefore part of a coordinated strategy.

In order to exploit the opportunity provided by the existence of the less transparent market (i.e., the auction) in period 1, the informed dealers optimally choose the quantity to trade in more transparent market in periods 1 and 2 and the bids at the auction. In particular, they maximize total profits (Π) jointly, in the secondary market in the two periods ($\Pi_1^i + \Pi_2^i$) and at the auction (Π_a^i). They solve:

$$\begin{aligned} Max_{x_1, x_a^i} \{\Pi\} &= Max_{x_1, x_a^i} \{\Pi_a^i + \Pi_1^i + \Pi_2^i\} = \\ &= Max_{x_1, x_a^i} \{E[V - P_1] X_1 + E[V - P_a] X_a^i + \Pi_2(P_1)\}, \end{aligned} \quad (4)$$

$$s.t. \quad : \quad Y_a = X_a^i + X_a^u + X_a^l - 1, Y_1 = X_1 + u_1, \text{ and } Y_2 = X_2 + u_2. \quad (5)$$

That is, the strategically account for the impact of their trades in the secondary market as well as for the impact of their bids at the auction. The benefits from trading in the secondary market are represented by the first term ($[V - P_1] X_1$),

while the benefits from trading at the auction are represented by the second term ($E[V - P_a] X_a^i$).

They display a trade-off for the informed dealer. He could try to immediately exploit his informational advantage by trading when the primary market is open. In this case, he would gain from trading in the secondary market. However, he would also give away most of his information and this would influence not only the price in the secondary market (through λ_1), but also the price in the primary market (through λ_a).

Alternatively, the informed dealer could behave strategically and exploit his informational advantage in the primary market. In this case, he would dissimulate his superior information when the primary market is still open, by reducing trade or manipulating prices -i.e., placing sell orders in the case he expect a high value of the bond. He would then bid aggressively at the auction and trade heavily in the afternoon on the secondary market when the primary market is closed and the information embedded in the auction price has not yet been revealed. In this case, he would derive most of his profits at the auction ($E[V - P_a] X_a^i$) and from trading in the afternoon ($\Pi_2(P_1)$).

The cost of this strategy would be to forfeit the gain that comes from the exploitation of his information in the secondary market. Intuitively, the choice between the two alternatives depends on the relative elasticity of the primary and secondary markets, that is, on whether informed dealers think that their trade has a higher impact in the primary market or in the secondary market.

The stronger the total demand at the auction (Y_a) and the stronger the price impact (λ_A), the more the price at the auction approaches the underlying value of the asset and the smaller is the possibility of profit for the informed dealer. This way of modelling the demand curve at the auction is reminiscent of the reduced form of the model of demand at the uniform auction derived by Wang and Zender (1999).

Finally, the *market makers* are the dealers who receive orders and set prices in the secondary market.

Assumption 3 Market clearing mechanisms.

Prices in the primary market are set through an auction mechanism. We assume that 1 unit of bond is auctioned off. The equilibrium price ("cut-off" price) at the auction is determined by the market clearing conditions.

In the secondary market, prices are set by a market maker. The market maker sets prices equal to the expected value of the asset, conditional upon the information available to him in the period. The market maker does not have any informational advantage. He starts the auction day ($t = 0$) with some priors about the terminal value of the bond (V), so that $V \sim N(P_0, \Sigma_0)$. Then he updates his beliefs by observing order flows in the morning and afternoon. Order flows (Y_t) are made of informed (X_t^i) and uninformed liquidity trading (u_t), for $t = 1, 2$. The update is a standard Bayesian rational update. In each period the signal (Y_t) is normally distributed, such that $Y_t \sim N(\mu_{Y_t}, \Sigma_{Y_t})$, where:

$$\mu_{Y_t} = E \left[\begin{array}{c} P_t \\ \alpha_t + \beta_t P_{t-1} \end{array} \right] \text{ and } \Sigma_{Y_t} = \begin{bmatrix} \Sigma_t & \beta_t \Sigma_t \\ \beta_t \Sigma_t & \beta_t^2 \Sigma_t + \Sigma_u \end{bmatrix}. \quad (6)$$

Theorem 1 *The market maker sets the price equal to the expected value of the asset on the basis of the information he derives from observing order flows. In particular: $P_t = E[V|Y_{t-1}, Y_t] = P_{t-1} + \frac{\beta_t \Sigma_{t-1}}{\Sigma_u^2 + \beta_t^2 \Sigma_{t-1}}(Y_t - \alpha_t - \beta_t P_{t-1}) = \mu_t + \lambda_t Y_t$, for $t = 1, 2$. (proof in Appendix).*

3.1 Equilibrium

We consider the joint equilibrium in the primary and secondary markets, under the assumption that informed dealers try to exploit their informational advantage by optimally choosing how much to trade in the secondary market in both periods and how much to bid at the auction in the first period. In order to do this, they solve a two-period optimization problem. Following the standard approach (Kyle 1985), we conjecture the existence a linear equilibrium in prices and linear demand functions at the auction and in the secondary market in both periods with price functions of the form:

$$P_a = \frac{1}{\beta^l} [X_a^i + X_a^u - 1], \quad P_1 = \mu_1 + \lambda_1 Y_1 \text{ and } P_2 = \mu_2 + \lambda_2 Y_2. \quad (7)$$

and demand functions for the informed ("i") trader and the speculative uninformed dealer ("u") equal to:

$$X_1^i = \alpha_1 + \beta_1 V, \quad X_2^i = \alpha_2 + \beta_2 V, \quad X_a^i = \alpha_a^i + \beta_a^i [EV|P_1] + \gamma_a^i V, \quad X_a^u = \alpha_a^u + \beta_a^u [EV|P_1], \quad (8)$$

where the subscripts "l", "s" and "a" refer, respectively, to the secondary market first period, secondary market second period and auction market.

We work by backward induction starting from the second period. Then, the decision of the informed dealer is conditional on *both* the value of the underlying asset (V) and the price in the first period (P_1).

Theorem 2 *Given the aforesaid assumptions, a sub-game perfect equilibrium exists in the secondary market in the second period, in which: $\lambda_2 = \frac{1}{2} \frac{\Sigma_1^{\frac{1}{2}}}{\Sigma_u}$, $\beta_2 = \frac{\Sigma_u}{\Sigma_1^{\frac{1}{2}}}$ and $\alpha_2 = \frac{-\mu_2}{2\lambda_2} = \frac{-P_1}{2\lambda_2} = -P_1\beta_2$. (proof in Appendix).*

That is, in period 2, the standard Kyle's results hold. Let's now move on to the first period.

Theorem 3 *Given the aforesaid assumptions, an equilibrium exists in the primary market, in which: $P_a = \frac{1}{\beta^l} (X_a^i + X_a^u - 1) = \frac{1}{2}V + E[V^u|P_1]$, where $X_a^u = \frac{5}{6}\beta^l E[V|P_1]$ and $X_a^i = 1 + \frac{1}{6}\beta^l E^u[V|P_1] + \frac{1}{2}\beta^l V$. (proof in Appendix).*

The demand in the primary market is fully determined in terms of uninformed dealers' expected value of asset ($E[V^u|P_1]$) and liquidity dealers' reaction to prices (β^l). The more the uninformed dealers are aware of informed dealers' trade, by directly observing the orders the informed dealers place with them and the prices in the secondary market before the auction, the more they will update upwards their expectation of the value of the bond. That is, uninformed investors' bidding in the primary market (X_a^u) is directly affected by their expectation of the value of the bond $E[V|P_1]$ and by the elasticity of the demand of the liquidity traders in the primary market (β^l).

Informed dealers' bidding (X_a^i) is instead affected by both the true fundamental value and uninformed investors' expectation of it. Alternative, it can be shown that informed dealers' bidding is a function of the true value and of forecasting errors made by the uninformed investors (i.e., $E^u[V|P_1] - V$). That is, the informed investors strategically condition their behavior in the primary market on the error the uninformed investors make.

Price in the primary market is therefore a function of the demand of both the informed and uninformed dealers and of the reaction function of the liquidity traders to prices. The more inelastic liquidity traders' demand is, the higher the price. We can interpret $\frac{1}{\beta^l}$ as the impact of the submitted bids on the price schedule and β^l as the equivalent of the market depth in the primary market.

We now move on to the first period. We solve the model parametrically in terms of two variables: uninformed investors' expected value of the bond ($E[V^u|P_1]$) and

liquidity dealers' reaction to prices (β^l). In particular, we define two variables: $\theta_\lambda = \frac{1}{\lambda_1 \beta^l}$ and $\theta_\beta = \frac{E^u[V|P_1]}{P_1}$.

The first variable (θ_λ) represents the ratio between the market depth in the secondary market in the first period ($\frac{1}{\lambda_1}$) and the market depth in the primary market (β^l). We can therefore define θ_λ as the parameter that relates the market depth in the primary and secondary market (i.e., $\frac{1}{\beta^l} = \theta_\lambda \lambda_1$). This also captures the *quantity* impact of operating in the primary market expressed as relative to the impact in the secondary market. Indeed, it can be intuitively thought of as the ratio between the market depth in the secondary market in the morning and the market depth in the primary market. Informed dealers can either trade in the secondary market before the auction, and in this case they will move prices by a measure (λ_1), or they will submit bids at the auction. In this case, they will affect prices by a measure ($\frac{1}{\beta^l}$).

The second parameter (θ_β) represents the relationship between price in the primary market and expected value of the asset for traders who submit bids in the primary market. The higher the value of θ_β , the more traders will rely on the level of prices in the secondary market when the primary market is open, in order to make forecasts to be used in the primary market. It represents the *learning* impact of the trade that takes place in the secondary market. By trading in the primary market when the primary market is open, the informed dealer affects prices and therefore reveals information to the less informed dealers, influencing their demands in the primary market.

The solution of the model will be parametric in terms of these two parameters. There are two reasons why we proceed in this way. First, the model would not be solvable otherwise, we just should resort to numerical estimation, with no possibility of carrying out any comparative statics. With this approach, on the contrary, we can see how the main results change for a whole gamut of values of θ_λ and θ_β . We will show that for all the plausible values of the parameters the results do not change. Therefore, even if we do not show what the value is at the equilibrium where both the primary and the secondary market simultaneously clear, still we are able to report all the possible equilibria at which the two markets may possibly clear. Moreover, we will show that at all these equilibria the results and the empirical restrictions go in the same direction.

The second reason is the additional degree of freedom we gain in the specification of the reaction function of the uninformed speculative dealers in the primary market.

Indeed, while we may reasonably assume that traders in the secondary market use a standard rational updating rule in order to infer the underlying value of the asset (V) by observing orders, the same cannot be said with equal confidence for the uninformed investors in the primary market. Indeed, these dealers are meant to represent not only market makers and dealers, but also institutional investors and other intermediaries who observe the total trade in the morning and, on the basis of them, they post bids in the primary market. We can assume that they may overreact or under-react to morning trading volume. Therefore, by specifying their reaction to morning trade parametrically we can see how the equilibria and the main results will change as we change the values of the reaction function, deviating from full rational learning.

At the same time, by considering all the possible plausible values, we are also implicitly solving for the equilibrium one. This is sufficient for our limited purpose of comparing equilibria when the two parallel markets are open ("auction" equilibria) to the equilibria when only the more transparent market is open ("no-auction equilibria") and deriving testable restrictions. The fact that we consider all the possible plausible values of the parameters and the fact that, for all of these values, the exercise of comparative statics provides the same results guarantees their robustness.

Theorem 4 *Given the aforesaid assumptions, a sub-game perfect equilibrium exists in the first period, in which: $\lambda_1 = \frac{\Gamma}{(1+\Gamma^2)} \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u}$, $\beta_1 = \frac{1}{\Gamma} \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}$ and $\Sigma_1 = \Sigma_0 \frac{1}{1+\Gamma^2}$, where $\Gamma = \sqrt{4w^2 - 1}$, and $w = \frac{\lambda_2}{\lambda_1}$ is the solution to the cubic equation $(-8\theta_\lambda + 4\theta_\beta)w^3 + 4\theta_\lambda w^2 + (4\theta_\lambda - \theta_\beta^2)w - \theta_\lambda = 0$ (proof in the Appendix).*

This also implies that: $\beta_2 = \frac{2k}{\Gamma} \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}$ and $\lambda_2 = \frac{1}{(1+\Gamma^2)} \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u}$. Given that the equilibrium is defined in terms of the parameters θ_λ and θ_β , we can define the no-auction equilibrium as the one where $\theta_\beta = 0$ and $\theta_\lambda = 1$. That is, if uninformed speculative dealers' bids in the primary market are not affected by the price in the secondary market and the price impact of the informed dealers in the two markets is identical, the existence of the primary market will not affect informed dealers' behavior in the secondary and the equilibrium values in the secondary market will be the same as those in the case the primary market did not exist.

Corollary 1 *Given the aforesaid assumptions, in the case the primary market did not exist, the equilibrium in the secondary market would be characterized by: $\beta_1^{NA} =$*

$0.6671 \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}$, $\beta_2^{NA} = 1.2021 \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}$, $\Sigma_1^{NA} = \Sigma_0 \frac{1}{1 + \beta_1^2 \frac{\Sigma_0}{\Sigma_u}} = 0.6920 \Sigma_0$, $\lambda_1^{NA} = 0.4617 \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u}$ and $\lambda_2^{NA} = 0.4159 \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u}$. (*proof in the Appendix*).

We will consider this equilibrium as the benchmark case, that is the "non-auction" case. It is worth noticing that these equilibrium values are *the same as the ones we would get by solving a two period problem with no primary market*.

We will now compare the case when the two parallel markets are simultaneously open ("auction case") to the case when the primary market is closed and only the more transparent market is open ("non-auction case"). We can now this by comparing the two equilibria for different values of the two parameters we have solved for - i.e. θ_λ and θ_β . We will use the Figures 1-4 where we report some of results of comparative statics.¹¹ This delivers some empirically testable restrictions that can be directly brought to the data. The restrictions can be divided in two types: the relationship between prices and informed dealers' behavior in the two markets (both the primary and the secondary markets) and the effects of the auction on the depth of the secondary market.

3.2 Empirical implications

3.2.1 Prices and dealers' strategies

Informed dealers try to hide their information in the more transparent market, in order to better exploit it in the less transparent one. This implies that informed dealers' trading in the secondary market when the primary market is open (auction days) is lower than during the period when the primary market is close (non auction days). In particular, during auction days informed dealer's trading is lower in the morning than in the afternoon.

Depending on the underlying parameters (β^l , θ_β , Σ_u and Σ_0), hiding can turn into price manipulation. In this case the model predicts that informed dealers sell the bond in the morning ($X_1^i < 0$), buy it back in the afternoon ($X_2^i > 0$) and bid aggressively in the primary market ($X_a^i > 0$).

Manipulation is positively related to the learning impact (θ_β) and negatively related to the quantity impact (θ_λ). That is, the more informed dealers' trade in

¹¹We display the case where $\Sigma_0^{\frac{1}{2}} = \sigma_u$. The other cases - i.e. $\Sigma_0^{\frac{1}{2}} \geq \sigma_u$ - provide analogous results and are not reported but are available upon request from the authors.

the secondary market affects uninformed investors' demand in the primary market, the more informed investors will sell in the secondary market in order to affect uninformed investors' beliefs. This means that the possibility of manipulation increases with uninformed investors' degree of "overreaction" to the prices in the more transparent market (i.e., higher θ_β).

At the same time, informed investors will be more induced to manipulate when the benefits they derive from such a strategy are high, i.e. when they can use their informational advantage in the primary market without affecting the price at the auction too much. That is, the quantity impact in the primary market is low. This is the case when liquidity traders' demand elasticity in the primary market is low and the market depth in the primary market is high (i.e., lower θ_β and higher β^l). Therefore, informed dealers will concentrate in the primary market and to manipulate prices in the secondary market, if they can safely assume that liquidity traders will in any case provide liquidity with a bidding schedule relatively price-insensitive.

A graphical analyses of these intuitions is provided the Figures 1 and 2. They report the equilibrium values of informed dealers' demand in the primary and secondary market in the two periods (respectively X_A , X_1 and X_2) for different values of the price impact in the primary market (θ_λ) and of the learning impact in the secondary market (θ_β).

Figure 1 reports the equilibrium values computed assuming $\Sigma_u = \Sigma_0 = 0.5$, while Figure 2 reports the equilibrium values computed assuming $\Sigma_u = \Sigma_0 = 5$. It is interesting to note that informational uncertainty in the market - i.e. high volatility of liquidity trading (Σ_u) or higher uncertainty of market makers and uninformed investors' priors (Σ_0) - reduces manipulation. This is due to the fact that higher uncertainty makes it more difficult for the uninformed investors (who only observe prices) to figure out whether a reduction in prices is due to informed dealers' trades or temporary liquidity fluctuations. This muffles the effect of the learning impact, reducing the incentive to manipulate in the secondary market.

The empirical estimation will tell apart the pure hiding strategy from the manipulation one. In both strategies dealers drastically reduce trade (or even short in the case of manipulation) in the morning and increase it in the afternoon. The difference between the orders placed by the informed dealers in the morning and the orders placed in the afternoon period is "unusually" strong during auction days as compared to the difference in trading that takes place during the analogous period

in the non auction days. In particular, the change in informed dealers' trade from the morning to the afternoon is always stronger in auction days than in non-auction days.

The change in the pattern of trade is mirrored by a change in the price path. The price changes between the two periods are unusually strong during auction days and behave similarly to the changes in trades. In particular, if we define $\Delta_p = (P_2 - P_1)$ and $\Delta_x = (X_2^i - X_1^i)$, we have:

$$\frac{\partial \Delta_p}{\partial \theta_\lambda} < 0, \frac{\partial \Delta_p}{\partial \theta_\beta} > 0 \text{ and } \frac{\partial \Delta_x}{\partial \theta_\lambda} < 0, \frac{\partial \Delta_x}{\partial \theta_\beta} > 0. \quad (9)$$

This implies a positive correlation between the change in prices between the two periods (Δ_p) and the change in trading activity of the informed dealers (Δ_x) in the same interval. That is,

$$\text{corr} [\Delta_p, (X_2^i - X_1^i)] > 0. \quad (10)$$

It can be also shown that a positive correlation is predicated between informed dealers' demand at the auction and price changes in the secondary market between the two periods. That is, the price movements in the afternoon and around result announcement time should be positively correlated to the shares of submitted bids of the informed dealers, such that:

$$\text{corr} [\Delta_p, X_a^i] > 0. \quad (11)$$

3.2.2 Market depth and dealers' strategies

The previous results deliver a set of empirical restrictions that relate the market depth of the auction days to the depth of the non-auction days. In particular, it can be shown that:

$$\lambda_1^{NA} > \lambda_1, \lambda_2^{NA} < \lambda_2, \quad (12)$$

that is, market depth in the morning is higher during auction days than during non-auction days. Conversely, market depth in the afternoon is higher during non-auction days.

Figures 3 and 4 report the market depth in the secondary market in the morning and afternoon for different values of θ_λ and θ_β . It immediately appears that the existence of the auction increases market depth in the morning and reduces it in the afternoon.

An increase of the learning impact raises market depth before the auction and reduces it after the auction. Indeed, the more the uninformed dealers use the prices

in the secondary market to decide their bids at the auction (i.e. the higher θ_β), the more the informed dealers react by reducing the information driven trade. This is also confirmed by the analysis of informed dealers' demand in the two periods. As θ_β increases, informed dealers' demand in the secondary market drops in the morning and increases in the afternoon. This reduces the amount of information the market makers are able to infer from trades and therefore lowers the sensitivity of prices to informed trade (λ_1) and increases market depth. Therefore, strategic behavior reduces price sensitivity. The opposite holds for the afternoon. Once the deadline for the submission of bids at the auction is over, informed dealers can increase their informationally related trade with no worry of repercussion on the auction price. This increases the amount of information the market makers are able to infer from trades, and therefore increases the sensitivity of prices to informed trade (λ_2) and reduces market depth.

Conversely, an increase of the quantity impact (θ_λ) reduces market depth before the auction and increases it after the auction. Indeed, if the informed dealer affects prices more at the auction than in the secondary market, he will be more willing to exploit his informational advantage in the secondary market than at the auction. This implies that the higher θ_λ is, the more the informed dealer increases his information-motivated trade in the secondary market. This is also confirmed by the analysis of informed dealers' demand in the two periods. As θ_λ increases, informed dealers' demand in the secondary market increases in the morning and drops in the afternoon. This increases the amount of information the market makers are able to infer from trades, and therefore increases the sensitivity of prices to informed trade (λ_1) and reduces market depth. The opposite is true in the afternoon, when the auction deadline is over.

Analogous reasoning explains the relationship between the quantity and information effects and informed dealers' reaction function (β) in the two periods. In particular, an increase in the quantity impact (θ_λ) increases both β_1 and β_2 , while an increase in the information impact (θ_β) decreases both β_1 and β_2 . Indeed, the more the informed dealer affects the cut-off price at the auction, the less he will bid at the auction, focusing on the secondary market. This means that his reaction function in the secondary market will be higher. In other words, a change in the value of the bond (V) will induce a stronger demand in the secondary market (i.e., higher β). The opposite happens in the case of an increase of the learning impact. A higher impact induces the informed dealer to try to exploit his informational advantage

mostly at the auction. This muffles his reaction in the secondary market.

4 The datasets and variables

The primary market dataset contains the demand schedules submitted by the individual intermediaries, the number of bonds they have been allotted at the auction and the auction clearing price. The dataset is accurate in terms of capturing the exact timing of the events, as it covers the period when bonds are issued through a screen-based computerized auction mechanism.¹²

We will focus only on the re-openings, that is those auctions where existing bond are auctioned off. These represent 90% of the auctions. For each auction, we consider the total number of both submitted and winning bids. Given that prices on the secondary market are quoted net of transaction fees and accrued interests, we deduct transaction fees and accrued interests from the auction prices.¹³ The fact that we focus only on the auctions relative to the issues of securities already traded in the secondary market,¹⁴ allows us to control for squeeze and to better concentrate on the effects of asymmetric information. Given that the bond is already traded in the secondary market, it is more difficult for a squeeze to materialize.

We use the R.N.I. announcement time as the time of the release of information to the market. The sample of auctions considered is further restricted to the set of auctions where the Treasury announces the results before 4.30 p.m.. We thus have enough time to study the transactions in the secondary market after the time of the announcement of the auction results.

The secondary market dataset contains for all the listed bonds, the time at which the transaction is executed, the size of the transaction, the price and the name of the counterparts. The dataset contains all the transactions from 29 September 1994 to 28 February 1996 for all the listed bonds (total of 37). In all, they total 1,393,437 transactions. For each transaction, we have available the time at which the transaction is executed, the size of the transaction, the price and the name of

¹²Also, the time interval between the submission of the demand schedules to the Treasury and the time of the release of the auction results is very short.

¹³In particular, the Treasury forfeits a day of accrued interest to the purchaser. Therefore, the price actually paid is equal to the official auction clearing price multiplied by $(1-1/365)$. Given that the price on the secondary market is net of transaction fees, we also deducted the commission fee paid by the dealers that are variable for the type of bond and period.

¹⁴Indeed, in the case of the first issue of a new bond, there is no proper secondary market on which dealers can trade (see before).

the counterparts and the identification of the dealer who originated the transaction. This allows us to identify the trades that have been originated by the market makers from those he has received from other dealers.

Around each auction, we create time series of transaction prices on the secondary market for the specific bond being auctioned off. They are constructed by using the actual transaction prices at the beginning and at the end of each sub-period. In particular, prices on the secondary market are defined in relation to the different time intervals: $P_{t,B}$ is the price of the last transaction in the bond before the deadline to submit bids at the auction (1.00 p.m.), while $P_{t,A}$ is the price recorded immediately after it. $P_{t+1,B}$ is the price recorded immediately before the auction result announcement ($t+1$), and $P_{t+1,A}$ is the price recorded immediately after the auction results have been announced. Also, we define the auction price as P_W .

Dealers are ranked according to their behavior in the primary market. The share in the primary market of the first n dealers is determined by calculating the quantity that each dealer gets allotted at the auction. The ranking is done for each auction. We consider the biggest intermediaries as the most likely candidates to be the informed dealers, under the hypothesis that their dominant market share allows them to be better informed. This assumption is supported by the recent theory that defines "non-payoff relevant information" (Cao and Lyons, 1999). In particular, the temporary unbalance between demand and supply, due to liquidity shocks and temporary price pressures, can be considered as private information for the dealers who intermediate the demand of the final investors. Indeed, by observing the demand schedules that are submitted to them, the dealers can make forecasts about market prices. The biggest dealers, those who intermediate a higher fraction of demand from the final investors, are therefore the ones most likely to be informed.

For such groupings, we calculate their demand at the auction and their trading on the secondary market. In particular, in the primary market we first aggregate the winning (submitted) bids of the groups constructed as above, and then we standardize them on the basis of the total amount of bonds allotted (demanded) at the auction.

We proceed analogously in the secondary market, considering the buy and sell orders originated by the groups constructed as above. Given that more information can be gleaned from the observation of the actual purchases and sales, we consider separately the purchases and sales. We standardize the buy (sell) orders of each groupings by dividing them for the total buy (sell) orders of the bond on auction

by the market as a whole in the analogous interval.¹⁵ The fact that we consider the transactions of the biggest dealers in relative terms with respect to the market allows us to separate price pressure effects from informational effects. While we can expect that a high trading volume of the biggest dealers can generate an excess demand that affects prices, no such effect can be attributed to a change in the *relative* trading volume of the biggest dealers. Indeed, the fact that a higher percentage of orders is originated by the biggest dealers should not affect prices unless these dealers have some specific characteristics - e.g., they are the most informed ones.¹⁶

We also create some variables that proxy for the dispersions of beliefs and/or information, such as the price volatility of the bond being traded in the secondary market and the dispersion (standard deviation) of the bids at the auction. These variables are designed to control for market-wide swings due, for instance, to exogenous shocks or public information arrival. We also use some control variables traditionally employed in the literature, such as the financial duration of the bond being auctioned off and the Herfindhal index of concentration of the bids at the auction.

4.1 Identification assumptions

The strategies of the dealers are designed in order to exploit the informational advantage they have. Therefore, in order to analyze such strategies, we need to identify the dealers who are informed and to construct a proxy for the information they try to exploit.

What is private information in the bond market? Unlike the equity market where information is mostly about fundamentals and/or stock-related idiosyncratic shocks, information in the bond market is mainly about liquidity shocks (Cao and Lyons, 1999). Foreknowledge of terminal investors' demand (i.e., banks, mutual funds, institutional investors, pension funds, insurance companies) provides an informational advantage. This advantage is particularly relevant in the case of a Treasury auction, when the demand schedule of the terminal investors defines the price of the bond at the end of the auction day.

¹⁵That is, if we define BD as the trading volume of the biggest dealers in the bonds on auction and AD as the trading volume of all the dealers in the bonds on auction, we consider: BD/AD .

¹⁶Also, the standardization helps us to control for the fact that the *during* period and the *after* period are variable, with a length depending on the time the Treasury announces the results. The standardization prevents the relationship between trading intensity and the change in prices around the auction from being due simply to the variable length.

In order to define the informed dealers, we take advantage of the richness of our dataset and use the primary market. We proceed as follows. First, for each dealer, we construct the profits they make at the auction. These profits are equal to the difference between the cut-off price at the auction and the price at the end of the day multiplied by the number of bonds they get allotted at the auction. Then, we rank all the dealers according to these profits. Then, we group dealers into 5 categories: the category made of the first 10 dealers (G10), the category made of the first 15 dealers (G15), the category made of the first 20 dealers (G20), the category made of the first 30 dealers (G30) and the category made of the dealers ranking between 50 and 100 (Others).¹⁷ The last category has been constructed as a residual class that should proxy for the less informed dealers - i.e., the dealers who generate the lower profits at the auction.

In order to construct a proxy for the information, we use the change in prices in the secondary market around the auction. In particular, we consider $\Delta P_t = P_{t+1,A} - P_{t,B}$ and $\Delta P_t = P_{t+1,A} - P_{t,A}$. This identification is standard in the empirical literature (Bikchandani, Edsparr and Huang, 1993, Simon, 1994) that considers the change in price around the auction as indicative of informational content of the auction. We further test this hypothesis by estimating:

$$\Delta P_t = \alpha + \beta I_t + \gamma D_t + \eta_t, \quad (13)$$

where I_t represents a measure of the additional information not contained in the market price in the secondary market, but part of the auction price. The null hypothesis that no additional information is included in the auction price would imply a value of β equal to zero. The alternative would assume a positive β in the case where the spread of private information relative to the auction takes place before the time of the announcement of the results. I_t is constructed as either the residuals of the regression of the auction price over the price in the secondary market at 1.00 p.m. (Bikchandani, Edsparr and Huang, 1993)¹⁸ or the difference between the price at the auction and the price in the secondary market at the deadline for the submission of bids at the auction.¹⁹ The idea is that I_t represents the change

¹⁷The criterion used to select the last class is aimed at constructing a meaningful residual class. We therefore skip 20 dealers (from 30 to 50) in order to identify dealers who have a significantly different behavior in terms of profits at the auction. At the same time, we restrict the cut-off to the first 100 dealers, as the remaining dealers trade very infrequently in the secondary market and rarely participate at the auction.

¹⁸In particular, $P_w = \alpha + \beta P_{t,B} + \varepsilon_t$.

¹⁹Simon (1994). In particular, we use both the difference between the price at the auction and

in price not forecastable, and can proxy for the unexpected information component of the auction.

The value of β is between 0.13 and 0.29, depending on the specification, with very significant *t-stat*.²⁰ This shows that there is some information that is revealed at the auction. This information gets impounded in prices part in the afternoon and part at the announcement of the results of auction. Therefore, the change in prices in the afternoon (ΔP_t), provides a good proxy for this information. This also suggests, in line with the analogous findings in the US, that some dealers may be able to exploit their private information.

Therefore, by looking at the correlation between dealers' strategies (trades and profits) and the change in prices we are in effect studying how dealers' strategies relate to the informational content of the auction. A high correlation suggests that some strategies are played at the very time when the informational content of the auction is higher and therefore the informational advantage of the informed dealers is greater.

Let's now move on to directly inspect dealers' strategies. We will look at the evidence of strategic trading, considering separately the different hypotheses described above.

5 Parallel markets and dealers' strategies

5.1 Empirical restrictions

Our model suggests that dealers act strategically in the more transparent market (secondary market) in order to be able to exploit their information in the more opaque one (primary market). In particular, dealers can dissimulate their information by refraining from trading before the auction, or they can try to directly manipulate prices in order to generate mistaken beliefs in the other dealers.

However, the dealers who are market makers (on average all the informed dealers are market makers) are also required to stay on the market, post bid and ask prices and trade a certain minimum fraction of orders. While in our model we separated

the price in the secondary market ($P_w - P_{t,B}$) before the deadline and the difference between the price at the auction and the price in the secondary market ($P_w - P_{t,A}$) after the deadline.

²⁰In particular, in the case where $\Delta P_t = P_{t+1,A} - P_{t,B}$, β is 0.15, 0.16 and 0.13 for specifications where I_t is equal to $(P_w - P_{t,B})$, $(P_w - P_{t,A})$ and ε_t from the regression $P_w = \alpha + \beta P_{t,B} + \varepsilon_t$. In the case where $\Delta P_t = P_{t+1,A} - P_{t,A}$, β is 0.26, 0.22 and 0.29 for specifications where I_t is equal to $(P_w - P_{t,B})$, $(P_w - P_{t,A})$ and ε_t from the regression $P_w = \alpha + \beta P_{t,B} + \varepsilon_t$.

market makers from informed dealers, in fact the same dealer is often both an informed dealer who places orders with other dealers and a market maker himself who receives orders by other dealers. We will therefore separately consider the trades that the dealer originates ("originated trades") and the trades the dealer receives ("received trades").

The observation of the originated trades allows us to directly infer their strategies. Indeed, we expect that if dealers have bullish expectations about bond prices and want to induce the other dealers to have bearish ones, they will sell in the interdealer market. The direct empirical implication is an increase in the originated sell orders of the more informed dealers and a reduction of their profits that ensue from such orders.

At the same time, these sell-orders, in order to be effective, have to take place at below market prices. This implies that the dealers who receive these orders will report an increase in the received trades at very favorable conditions. That is, an increase of received buys and an increase of the profits that ensue from such orders.

No change in either originated trade or received trade is instead expected in the absence of strategic behavior.

It is worth noticing that the dealer can use the possibility of placing orders with other dealers either to reconstitute his inventory or to enact his signal-jamming strategy. In order to be able to separate the two hypotheses we need to consider dealer's behavior in the primary market. In the case where the dealer resorts to placing orders with other dealers for inventory reasons, we would expect no correlation between such a behavior and the behavior in the primary market.

For example, let's assume that the dealer places sell orders with other dealers simply because his inventory has increased as a consequence of the receipt of many sell orders by other dealers. In this case, we do not expect that at the same time the dealer bids aggressively at the auction in order to get allotted the very same bonds he is trying to dispose of by selling to other dealers. Therefore, the correlation between his originated trades in the secondary market and his bids in the primary market should be either positive (i.e., he would bid aggressively at the auction only at the times when he is aggressively placing buy orders with other dealers in the secondary market) or zero (i.e., there would not be any correlation between the behavior of the dealer at the auction and his behavior in the secondary market).

The aforesaid strategies should also be reflected in the profits of the dealers. Indeed, the signal-jamming strategy may be very expensive as the dealers sell at

the very times when they expect prices to rise. We should therefore expect that dealers lose out the originated sell orders. At the same time, we would expect that the dealers who stand as counterparts to them profit out of the received orders.

The aforesaid implications give rise to a set of empirical restrictions that allow us to test for the existence of strategic behavior. We can recast them into the following set of restrictions on profits and trades at the time when the two parallel markets are open. This period is the morning. The restrictions relate informed dealers' behavior to the amount of private information they are endowed with. Strategic behavior should be more pronounced at the very times when the gains from private information are higher. Given that this information is proxied by the informational content of the auction - i.e., changes in prices after the auction has taken place - strategic behavior should be traceable at the very times when prices increase after the auction.

In particular, we define a set of restrictions jointly on received and originated trade as well on the profits that arise out of received and originated trades. We will consider both the primary and the secondary market.

Let's define Π_b^r and Π_s^r the profits that arise out of the trades originated by the dealers who place orders with other dealers, and Π_b^o and Π_s^o the profits that arise out of the orders that are received by the dealers. Let's also define T_b^o and T_s^o the trades that are originated by orders directly placed with other dealers and T_b^r and T_s^r the trades that are received by the dealers. The subscripts "b" and "s" refer, respectively, to "buy" and "sell".

For instance, Π_b^r are the profits due to transactions where the dealer has been required to buy - i.e., transactions executed at his "bid" price. Π_b^o are the profits due to transactions where the dealer has directly placed an order to buy with another dealer - transactions executed at the "ask" price of the other dealer he has placed the order with.

In the case of strategic behavior and price manipulation, we would expect:

$$\begin{cases} corr(T_s^o, \Delta P) > 0, \quad corr(T_b^r, \Delta P) > 0, \quad corr(T_b^{net}, B_a) < 0, \\ \quad \quad \quad corr(\Pi_s^o, \Delta P) < 0, \quad corr(\Pi_b^r, \Delta P) > 0. \end{cases} \quad (14)$$

In case of information dissimulation and no price manipulation, we would expect:

$$\begin{cases} corr(T_s^o, \Delta P) = 0, \quad corr(T_b^r, \Delta P) = 0, \quad corr(T_b^{net}, B_a) = 0, \\ \quad \quad \quad corr(\Pi_s^o, \Delta P) = 0, \quad corr(\Pi_b^r, \Delta P) = 0. \end{cases} \quad (15)$$

Finally, in case of no-strategic behavior, we would expect:

$$\begin{cases} \text{corr}(T_b^o, \Delta P) > 0, \text{corr}(T_b^r, \Delta P) > 0, \text{corr}(T_b^{net}, B_a) > 0, \\ \text{corr}(\Pi_b^o, \Delta P) > 0, \text{corr}(\Pi_b^r, \Delta P) > 0. \end{cases} \quad (16)$$

Restriction 14 defines dealer's strategic behavior. If the informed dealer manipulates prices in order to gain more advantageous conditions at the auction, he aggressively places sell orders with other dealers at the very times when the gains from his private information are higher ($\text{corr}(T_s^o, \Delta P) > 0$). This would result in losses ($\text{corr}(\Pi_s^o, \Delta P) < 0$). At the same time, if the number of informed dealers who play analogous strategies is high enough, this strategies would results in higher received purchases ($\text{corr}(T_b^r, \Delta P) > 0$) taking place at convenient prices and therefore, higher induced profits ($\text{corr}(\Pi_b^r, \Delta P) > 0$.) for the dealer who take the other side of this transactions. If the informed dealers do not change their bid-ask prices while the rest of the market widens the spread, the informed dealers will most likely be the ones taking the other sides of most transactions. However, this part of restriction 14 does not follow directly from the model.

Furthermore, in order for these correlation to suggest a strategic behavior and not just an inventory rebalancing one, we would expect the dealers to also bid aggressively at the auction. This would generate a correlation between purchases at the auction and overall net sales in the secondary market ($\text{corr}(T_b^{net}, B_a) < 0$).

Restriction 15 defines the strategic behavior of a dealer who merely wants to dissimulate his information, without manipulating prices in the opposite direction. In this case, we should not find any particular abnormal behavior at the times when prices increase after the auction. That is, both profits and trades received by a change in the bid-ask prices should not be correlated to prices (i.e., $\text{corr}(T_s^o, \Delta P) = 0$, $\text{corr}(T_b^r, \Delta P) = 0$, $\text{corr}(T_b^{net}, B_a) = 0$, $\text{corr}(\Pi_s^o, \Delta P) = 0$). Also, no correlation should exist between informed dealers' behavior in the primary market and secondary market (i.e., $\text{corr}(T_b^{net}, B_a) = 0$).

Finally, the other possibility is that the dealer may want to exploit his information advantage and increase his positions ahead of the auction, without being concerned about disseminating information. This is captured by restriction 16. In this case, the dealer would increase his purchases by both moving the bid-ask prices ($\text{corr}(T_b^r, \Delta P) > 0$) and directly placing orders with other dealers ($\text{corr}(T_b^o, \Delta P) > 0$), while at the same time bidding aggressively at the auction ($\text{corr}(T_b^{net}, B_a) > 0$). This would generate profits in the secondary market

($\text{corr}(\Pi_b^r, \Delta P) > 0$ and $\text{corr}(\Pi_b^o, \Delta P) > 0$), but would also reduce the possibility of profits in the primary market.

These restrictions can be directly brought to the data. Indeed, we can estimate profits and trades for all the dealers and see how they relate to our restrictions.

5.2 Preliminary evidence: informed dealers' trades and profits.

Evidence of strategic behavior can be inferred by looking at the trades and profits of the informed dealers and assessing whether they differ when both markets are simultaneously open from the other days. This implies comparing the days of the auction from the other non-auction days. We therefore separately consider the trades and profits of the informed dealers the days of the auction (respectively, T_{auct} and Π_{auct}) and the trades and profits on non auction days (respectively, T_{noauct} and Π_{noauct}).

The profits are constructed as in Madhavan and Smidt (1993). That is, the quantities purchased are multiplied by the difference between the price at the end of the day and the price at which the transaction is executed, while the quantities sold are multiplied by the difference between the price at which the transaction is executed and the price at the end of the day. Given that more information can be gleaned from the observation of the actual purchases and sales, we consider separately the purchases and sales.

The trades and profits on the non auction days are based on the transactions executed by the same informed dealers in the previous days for bond non on auction. In this case, for each auction we construct the series of buy and sells of the other bonds non being auctioned off at day t , in the previous 90 days (i.e., $t-1, t-2, \dots, t-89$) for the same time interval (i.e., morning). We separately consider the received trades and the originated trades and the profits arising out of them.

Dealers are grouped in 5 categories on the basis of the profits the dealers generate at the auction (G10, G15, G20, G30 and Others). The residual category (Others) proxies for the less informed dealers - i.e., the dealers who generate the lower profits at the auction.

In Table 2 we report the average values of trades and profits during the period for the different groups and the statistics (t -stat and p -value) of a t -test of differences between values for the auction days as opposed to the values for the non-auction days.

The results show that in general originated trades always provide higher profits than the received ones. This is consistent with standard microstructure theory that assumes that the market maker averages out the losses he incurs by trading with more informed dealers with the gains he generates by dealing with the less informed ones. In particular, the received trades on average generate losses. On the contrary, the profits arising out of originated trades for the informed dealers²¹ are on average positive due to the exploitation of the informational advantage. Indeed, the difference between received and originated profits is minimum in the case of the less informed dealers ("Others").

Also, there is a strong difference in informed dealers' behavior between the time when the two parallel markets are open (auction days) and the time when only the secondary market is open (non auction days). In particular, when the two markets are open, both originated and received trades are negative. That is, informed dealers reduce buying and starkly increase selling. This is less evident in the case of received trade, where the net trade is barely negative, and way more pronounced in the case of originated trade, where the net sells are of the same order of magnitude of the net buys when the primary market is closed (i.e., -3.40 versus 2.16 for the first 10 informed dealers). We will see later on that this is consistent with dealers' overall behavior where originated trade only depends on dealer's own strategy, while received trade also depends on other dealers' strategies.

Moreover, the selling strategy is expensive for the informed dealers. Indeed, while originated trades are always generating positive profits in the non auction days, as expected, they generate losses in the periods when both markets are simultaneously open (auction days). This seems to suggest that the informed dealers are willing to take short term losses in the morning in order to manipulate price and/or dissimulate their information advantage.

Finally, it is worth noting that, while the differences in both profits and trades are always statistically significant for any confidence interval in the case of the informed dealers, there does not seem to be a change in behavior for the less informed dealers. Indeed, in their case, the *t-test* fails to reject the null of no difference between auction and non action days.

These results provide evidence in favor of statistically significant differences in

²¹The dealer may also place orders with other dealers just to rebalance his inventory. However, this is less likely in the case of the more informed dealers. We will show by comparing informed dealers' behavior in *both* the primary and the secondary market that inventory rebalancing is not the main motive.

informed dealers behavior when the more opaque market is open. We will now examine this issue more in detail, by focusing on how dealers' strategic behavior is affected by their informational advantage (i.e., the informational content of the auction proxied by the changes in prices in the secondary market in the afternoon). We will separately consider trades and profits.

5.3 Strategies and dealers' trading behavior

Let's start by considering investors' trading behavior. We estimate the following specification:

$$\Delta P_t = \alpha + \beta B_{j,t} + \gamma S_{j,t} + \boldsymbol{\delta}' \mathbf{C}_t + \varepsilon_t, \quad (17)$$

where ΔP_t is the change in price at different time intervals for the bond being auctioned off in the primary market. We consider the two alternative specifications defined above (i.e., $\Delta P_t = P_{t+1,A} - P_{t,B}$ and $\Delta P_t = P_{t+1,A} - P_{t,A}$). $B_{j,t}$ and $S_{j,t}$ are, respectively, the buy and sell orders executed by the j th informed dealer. $\mathbf{C}$ represents a vector of control variables. They include the duration of the bond being auctioned off (*Duration*), the Herfindhal index of concentration of the bids in the primary market (*Herfindhal*), the cover ratio - i.e., the ratio between total demand and total amount of bonds allotted in the primary market - (*Dom/Allot*), the dispersion of both the submitted bids and the winning bids in the primary market (respectively *Volat Demand* and *Volat Allott*.) and the volatility in the secondary market (*Volat. Sec. Mkt*).

We consider both the case where $B_{j,t}$ and $S_{j,t}$ are the dealer-originated trades (i.e., the trades based on orders that the dealer places with other dealers), as well as the case where they are the dealer-received trades (i.e., the ones originated by other dealers placing orders with him).

We consider the different groups of dealers defined before: 4 groups of informed dealers (G10, G15, G20 and G30) and one residual group of uninformed ones (Others). We also look at the transactions executed in the afternoon. Indeed, while restrictions 14, 15 and 16 refer to dealers' strategic behavior at the time when both parallel markets are simultaneously open, it is nonetheless interesting to observe what happens after the less transparent market (i.e., the primary one) closes. The model would suggest that informed dealers revert their trades - i.e., place buy orders in the afternoon.

In order to control for market conditions, we also re-estimate equation 17 by using the trades by the same groupings in the previous 90 days at the same time (i.e.,

$t-1, t-2, \dots, t-89$) of the bond non on auction. These regressions are meant to assess whether any result we get by estimating equation 17 is due to the existence of the auction, or if it is due to the characteristics of the market we are considering. The results (nor reported)²² strongly support our hypothesis, as no significant relationship exists between changes in prices and informed dealers' behavior in days when the less transparent market is not open. This suggests that the type of strategy investors play is due to the temporary co-existence of two parallel markets characterized by different degree of transparency.

Equation 17 is estimated through a pooled regression, with a White's robust variance-covariance matrix. Let's start by considering the dealer-originated morning trades. Restrictions 14, 15 and 16, imply:

$$\begin{cases} \gamma > 0, & \text{in the case of price manipulation} \\ \gamma = 0, & \text{in the case of information dissimulation} \\ \gamma < 0, & \text{in the case of no strategic behavior.} \end{cases}$$

The results are reported in Table 4. Panel A reports the results for morning trades and Panel B reports the results for afternoon trades.

The results broadly support our working hypothesis. Indeed, there is a strong positive correlation between originated sales in the morning and changes in prices. This holds for both definitions of changes of prices. Furthermore, while this holds for the different groupings of informed dealers - i.e., the dealers who generate the highest profits in the primary market -, there is no correlation between the change in prices and the trading behavior of the less informed dealers ("Others"). Also, it is worth remembering that no analogous correlation exists for the case of trades on bonds on non auctions and their relative prices. Finally, while there is a negative correlation between sales and changes in prices, there is not any relationship between change in prices and purchases. This is consistent with the restriction 14.

In the afternoon dealers' behavior changes. In particular, there is a strong positive correlation between change in prices and dealer-originated purchases. This would suggest that dealers restock themselves in the secondary market once the primary market is closed as they are not afraid of affecting competitors' behavior any longer.

These results provide evidence in favor of price manipulation, or restriction 14.

²²They are available upon request from the authors.

Let's now consider the morning received trades. Restrictions 14, 15 and 16, imply:

$$\left\{ \begin{array}{l} \beta > 0, \text{ in the case of price manipulation} \\ \beta = 0, \text{ in the case of information dissimulation} \\ \beta > 0, \text{ in the case of no strategic behavior.} \end{array} \right.$$

Here the restrictions are a looser indication of dealers' strategies. Indeed, dealers' behavior is affected by dealers' own strategy (i.e., setting of bid-ask prices) as well as other dealers' strategies (i.e., direct placement of orders in the interdealer market). Our intuition is that, if the informed dealers do not change their bid-ask prices while the rest of the market widens the spread, the informed dealers will most likely be the ones taking the other sides of most (sell) transactions. This would induce a positive correlation between received sell orders ("received purchases") and change of prices.

The results are reported in Table 4, Panel C for morning trades and Panel D for afternoon trades. The results show a positive correlation between purchases and the change in prices in the secondary market once the primary market is closed. No relationship exists, instead, for the sales. This holds for both definitions of changes of prices. Also, while the relationship is significant and robust for the different groupings of informed dealers, there is no relationship in the case of the less informed dealers ("Others").

Furthermore, it is worth stressing that, as in the previous case, this positive correlation does not exist for bond non on auctions. This provides additional evidence in favor of the impact produced by the existence of parallel markets. Also, it is interesting to notice that in the afternoon the behavior changes and there is now a significant negative relationship between originated sales and change in prices.

However, we still have to separate inventory rebalancing from information exploitation motives. In order to do this, we have to consider dealers' behavior in the primary market.

5.4 Trading strategies in the two markets

The decision of the dealers to place sell orders aggressively with other dealers may be also due to a desire to rebalance their inventory positions at the very time when they receive many sell orders. To tell apart inventory rebalancing from strategic exploitation of the information advantage, we have to consider dealers' behavior in

the primary market. We therefore estimate:

$$N_{j,t} = \alpha + \beta Q_{j,t} + \gamma' \mathbf{C}_t + \varepsilon_t, \quad (18)$$

where $N_{j,t}$ is the net trade in the secondary market of the j th dealer, $Q_{j,t}$ is the quantity demanded at the auction by the j th dealer and $\mathbf{C}_t$ is a set of control variable as defined above. Analogously to the previous specification, we consider 5 groupings of dealers. The net trade ($N_{j,t}$) is constructed as the difference between buy and sell orders placed with other dealers. Alternatively, we consider the change in net trade between the afternoon and the morning. That is the difference between the net trade (buy orders less sell orders) in the afternoon and the net trade in the morning. Equation 18 is estimated through a pooled regression, with a White's robust variance-covariance matrix.

If dealers place orders with other dealers for inventory-rebalancing reasons, we would expect zero (or positive) correlation between net trade in the secondary market and demand in the primary market. Conversely, we would expect a negative correlation in the case of strategic behavior. This implies a negative correlation between net trade in the morning and demand and the auction and a positive correlation between the change in net trade between the morning and the afternoon and the demand at the auction. That is, we expect:

$\beta < 0$, where $N_{j,t}$ is net trade and $\beta > 0$, where $N_{j,t}$ is the change in net trade, in the case of strategic behavior and

$\beta = 0$, where $N_{j,t}$ is net trade and $\beta = 0$, where $N_{j,t}$ is the change in net trade, in the case of inventory rebalancing.

The results are reported in Table 5, Panel A for the case of Net trade in the morning and Panel B for the case of Change in trade between afternoon and morning. They support the working hypothesis. Indeed, a strong negative correlation exists between the demand in the primary market and the net trade in the secondary market. Also the correlation between the change in net trade and the demand in the primary market is positive and statistically significant. Moreover, the less informed dealers (Others) display an opposite behavior.

We have therefore evidence that informed dealers manipulate prices. This strongly supports our model. However, we still do not know whether the informed dealers are incurring short term (morning) losses to enact these strategies. We therefore turn to the analysis of their profits.

5.5 Strategies and dealers' profits

Let's now consider profits. In particular, we want to assess whether there is a systematic relationship between profits and changes in prices. We therefore estimate:

$$\Delta P_t = \alpha + \beta \Pi_{b,j,t} + \gamma \Pi_{s,j,t} + \boldsymbol{\delta}' \mathbf{C}_t + \varepsilon_t, \quad (19)$$

where ΔP_t is the change in price at different time intervals for the bond being auctioned off in the primary market. As before, we consider as alternative specifications $\Delta P_t = P_{t+1,A} - P_{t,B}$ and $\Delta P_t = P_{t+1,A} - P_{t,A}$. $\Pi_{b,j,t}$ and $\Pi_{s,j,t}$ are the profits of the j th dealer due, respectively, to buy and sell orders. $\mathbf{C}_t$ represents a vector of control variables, defined as in the specification 17.

Analogously to specification 17, we consider both the case of dealer-received trades (i.e., the ones originated by other dealers placing orders with him) as well the case of dealer-originated trades (i.e., the trades based on orders that the dealer places with other dealers).

Also, as before, dealers are grouped in 5 categories on the basis of the profits they generate at the auction (G10, G15, G20, G30 and Others). The residual category (Others) proxies for the less informed dealers. We consider both the transactions executed in the morning as well as the transactions executed in the afternoon. Equation 19 is estimated through a pooled regression, with a White's robust variance-covariance matrix.

We also re-estimate equation 19 by using the profits that the same classes of dealers make by trading on bonds non on auction in the previous 90 days at the same time (i.e., morning and afternoon). The results (not reported) confirm the previous findings and show that no significant relationship exists for the bonds non on auction. This further supports the existence of a close relationship between investors' strategies and the existence of two parallel markets characterized by different degree of transparency.

Let's start by considering the morning profits arising out of the trades the dealers originate by directly placing orders with other dealers. Restrictions 14, 15 and 16, imply:

$$\left\{ \begin{array}{l} \gamma < 0, \text{ in the case of price manipulation} \\ \gamma = 0, \text{ in the case of information dissimulation} \\ \beta > 0, \text{ in the case of no strategic behavior.} \end{array} \right.$$

The results are reported in Table 6. As before, Panel A reports the results for the profits arising out of morning trades and Panel B reports the results for the

profits arising out of afternoon trades. The results strongly support our hypothesis. Indeed, there is a significant and negative correlation between the change in prices in the secondary market and the profits arising out of the originated sales. That is, informed dealers place orders at non convenient prices and, in doing so, they incur losses. No relationship exists, instead, for the purchases. These results hold for both definitions of changes of prices and are significant and robust for the different groupings of informed dealers.

It is interesting to notice that the less informed dealers ("Others") are instead making significant profits in the morning. No analogous behavior exists for the case of bonds non on auction on non auction days. This is consistent with the previous results that show that dealers sell more in the morning when they expect prices to rise after the auction. It indicates a strategic behavior induced by the existence of parallel markets.

In the afternoon, the relationship between profits due to originated sales and change in prices becomes positive. That is, informed dealers, buy when their trade will not affect competitors behavior in the primary market. The profits that arise out of these buy orders are directly related to the degree of informational advantage (i.e., changes in prices). The more the informed dealers are confident that prices will rise the more they will buy. In the afternoon the less informed dealers' profits drop. Indeed, either there is no correlation between change in prices and profits or there is a negative correlation, suggesting the lack of information of such a class of dealers.

It is worth noticing that the change in prices in the afternoon may be induced by the very trading activity of the informed dealers themselves. In order to check for this, we also estimate the relationship between the change in prices around the time the results of the auction are announced ($P_{t+1,A} - P_{t+1,B}$) and informed dealers' trade in the afternoon. The results (not reported) show a statistically significant positive correlation. If the change in prices were merely a reaction to informed dealers's trade, we would expect to find no correlation between the news due to the announcement of the auction results and informed dealers' trade. In fact, there should not be any news at all as the information should have been directly impounded in prices by dealers' trade. The fact that this is not the case suggests that informed dealers increase their buys at the very time they are more confident about the information they have.

Let's now consider the profits the dealers derive from the trades they execute

when they receive an order (i.e., dealer-received trades). Restrictions 14, 15 and 16, imply:

$$\begin{cases} \beta > 0, \text{ in the case of price manipulation} \\ \beta = 0, \text{ in the case of information dissimulation} \\ \beta > 0, \text{ in the case of no strategic behavior.} \end{cases}$$

Indeed, the mere fact that some informed dealers try to influence prices by selling at below-market prices generates profits for the dealers who are on the receiving side of these transactions. These "receiving" dealers would display higher purchases generated by orders of other dealers and corresponding greater profits.

The results are reported in Table 6, Panel C for the profits arising out of morning trades and Panel D for the profits arising out of afternoon trades. The results show a statistically significant positive correlation between change in prices and the profits due to buy orders. No correlation exists between the change in prices and the profits due sell orders. This holds for both definitions of changes of prices as well as for the different groupings of informed dealers (G10, G15, G20, G30). As in the previous case, the correlation between the change in prices and the trading behavior of the less informed dealers (Others) is either zero or negative. In the afternoon we find a negative correlation between change in prices and profits generated by dealer-received purchases.²³

As a further test, let's consider the relationship between dealers' net profits and changes in prices. That is we estimate:

$$\Delta P_t = \alpha + \beta \Pi_{ove,j,t} + \delta' \mathbf{C}_t + \varepsilon_t, \quad (20)$$

where $\Pi_{ove,j,t}$ are the overall profits constructed as the sum of the profits arising out of originated trades and of the profits arising out of received trades. The variables ΔP_t and $\mathbf{C}_t$ are defined as before. If informed dealers are acting strategically we expect:

$$\beta < 0 \text{ in the morning and } \beta > 0 \text{ in the afternoon.}$$

The results, reported in Table 6, Panel E and F, strongly support the working hypothesis. Indeed, there is a significant negative correlation between changes of prices and profits in the morning and a positive correlation between them in the afternoon. This holds for the different groupings of informed dealers (G10, G15,

²³This finding is difficult to interpret, as informed dealers incur losses with the received transactions in the afternoon, at the time when they are actually making profits out of the originated trades.

G20, G30). The uninformed dealers (Others) display an opposite behavior. The less informed dealers seem to take advantage out of informed dealers' strategic behavior in the morning, by increasing their profits at the very time when the informational advantage of the informed dealers is greater (i.e., bigger changes in prices). In the afternoon, instead, when the strategic incentive to manipulate prices disappears, the informed dealers fully exploit their informational advantage by increasing their profits at the time when information is higher. This generates losses for the less informed dealers (i.e., negative correlation between change in prices and profits).

Overall, the empirical findings from the tests of restrictions 14, 15 and 16 support the hypothesis of strategic behavior and, in particular, of price manipulation. It seems that informed dealers' strategy is based on selling at a loss in the more transparent market (secondary market) in the morning, when both market are open, aggressive bidding in the less transparent market (primary market).

It is worth adding that we also considered an "institutional" classification. That is, we constructed groupings for the specialists and groupings for the ordinary market makers. However, no statistically significant pattern emerges in this case. Only ordinary market makers's behavior seems to be related to changes in prices in the secondary market. In particular, only a sub-group of the ordinary market makers seems to be the important (not reported in tables). No relationship exists for specialists in general.²⁴

Furthermore, there does not seem to be a close relationship between strategic behavior in the primary market and institutional classification. This confirms our information-based story as strategic behavior and the relationship between trade and price is not related to dealers' institutional characteristics, but to their degree of informativeness.

We will now endeavor to take a step forward, by considering how the opening of a less transparent market impacts the market depth of the more transparent one.

6 Parallel markets on market depth

The temporary opening of a parallel market affects the market depth of the existing markets. The null hypothesis is that there should not be any significant change in market depth when the less transparent market is open (auction days). The

²⁴This is coherent with the observation that the institutional requirements - i.e. minimum trading in the secondary market, participation at the auction - influence specialists' strategies.

alternative, embodied in equation 12, assumes that market depth is higher than average when the less transparent market is open. That is, in auction days market depth in the secondary market is higher than average in the morning and is lower than average in the afternoon.

In order to test this hypothesis, we estimate the market depth as suggested by Glosten and Harris (1988). This method relates the change in prices at transaction time with the inverse of the market depth as derived in a Kyle-type model. In particular, we estimate:

$$\Delta p_t = \lambda q_t + \psi Z_t + \varepsilon_t \quad (21)$$

where Δp_t is the change in price at time t , q_t is the signed order flow at time t and Z_t denotes the sign of the incoming order at time t (+1 for a buyer-initiated trade and -1 for a seller-initiated trade). The error is assumed to be white noise. We estimate equation 21 for the morning and the afternoon separately for each auction day and for the four non-auction days preceding each auction. For the non-auction days we consider the same time window that was used for the auction that went off the particular week ²⁵ A *t-test* is performed to gauge the statistical significance of the difference between the auction and non-auction mean values of λ .²⁶

The restrictions embedded in equation 12 are:

$$\lambda_{mor}^{noauct} > \lambda_{mor}^{auct}, \lambda_{aft}^{noauct} < \lambda_{aft}^{auct} \quad (22)$$

$$\lambda_{mor}^{noauct} > \lambda_{aft}^{noauct}, \lambda_{mor}^{auct} < \lambda_{aft}^{auct}. \quad (23)$$

That is, market depth is higher than average in auction days in the morning and is lower than average in the auction days in the afternoon (restriction 22) and during auction days market depth is higher in the morning than in the afternoon, while during non-auction days market depth is higher in the afternoon than in the morning (restriction 23).

We have different ways of defining the "average". Indeed, we can consider as average the market depth of the other bonds that are not on auction ($\lambda_{auct,mor}^{noauct}$ and $\lambda_{auct,aft}^{noauct}$) as well as the market depth of the bond that is on auction in the days before the auction ($\lambda_{t,mor}^{auct}$ and $\lambda_{t,aft}^{auct}$).

²⁵We select these days in order to control for outside exogenous factors which could affect market depth. Therefore we limit ourselves only to the days within the same week.

²⁶A *t-test* is performed to gauge the statistical significance of the difference among the auction and non-auction mean and median values for λ . Given that the distributions of the resulting parameters (λ) do not have the same variance, we use a t-test for unequal variance.

In order to test restriction 22, we can compare the market depth for the bond on auction ("auct") in the morning of the auction day ($\lambda_{auct,mor}^{auct}$) with the depth for the bond on auction during the morning of the previous days ($\lambda_{t,mor}^{auct}$) as well as the depth for the bond on auction in the afternoon of the auction day ($\lambda_{auct,aft}^{auct}$) with the depth for the bond on auction during the afternoon of the previous days ($\lambda_{t,aft}^{auct}$).

In order to test restriction 23, we can directly compare the morning and afternoon market depths for the bond on auction for both auction ("auct") and non-auction days ($\lambda_{t,aft}^{auct}$ and $\lambda_{t,mor}^{auct}$) and the morning and afternoon market depths for the bond non on auction ("noauct") for both auction and non-auction days ($\lambda_{t,aft}^{noauct}$ and $\lambda_{t,mor}^{noauct}$).

The results of these comparisons are reported in Table 7. All the results agree, regardless of the alternative specifications we use to define the average market conditions. It appears that market depth is always higher than average in auction days in the morning and is lower than average in the auction days in the afternoon (Panel A). Also, during auction days ("same day") market depth is higher in the morning than in the afternoon, while during non-auction days ("Auct.Day-t") market depth is higher in the afternoon than in the morning (Panel B). There seems to be no difference between afternoon and morning for the bond on auction on non auction days, while the difference is always significant and negative - as expected - for the bond non on auction, both for auction days and non auction days.

As additional robustness check, we compare the differences in market depths between morning and afternoon of the bond on auction with the differences in market depths between morning and afternoon of the bonds non on auction. In particular, we compare the differences between the depths in the morning and in the afternoon of the bond on auction the auction day ($\lambda_{auct,aft}^{auct} - \lambda_{auct,mor}^{auct}$) with the differences between the depths in the morning and in the afternoon of the bond non on auction the auction day ($\lambda_{auct,aft}^{noauct} - \lambda_{auct,mor}^{noauct}$), as well as the differences between the depths in the morning and in the afternoon of the bond on auction in the days before the auction day ($\lambda_{t,aft}^{auct} - \lambda_{t,mor}^{auct}$) with the differences between the depths in the morning and in the afternoon of the bond non on auction in the days before the auction day ($\lambda_{t,aft}^{noauct} - \lambda_{t,mor}^{noauct}$). These results are displayed in Panel C.

We also compare the differences between the depths in the morning and in the afternoon of the bond on auction the auction day ($\lambda_{auct,aft}^{auct} - \lambda_{auct,mor}^{auct}$) with the differences between the depths in the morning and in the afternoon of the bond on auction in the days before the auction day ($\lambda_{t,aft}^{auct} - \lambda_{t,mor}^{auct}$), as well as the differences

between the depths in the morning and in the afternoon of the bond on non auction the auction day ($\lambda_{auct,aft}^{nonauct} - \lambda_{auct,mor}^{nonauct}$) with the differences between the depths in the morning and in the afternoon of the bond non on auction in the days before the auction day ($\lambda_{t,aft}^{nonauct} - \lambda_{t,mor}^{nonauct}$). These results are displayed in Panel D.

Also in this case, the results agree and show that regardless of the alternative specifications we use to define the standard market conditions, market depth is always higher than average in auction days in the morning and is lower than average in the auction days in the afternoon.

This suggests that the existence of a parallel market characterized by a different degree of transparency may significantly affect the degree of liquidity of the existing market in a non trivial way.

7 Conclusions

We studied how dealers behave when there exist parallel markets for the same asset characterized by very different degrees of transparency.

We identified the optimal trading strategy and argued that, depending on the features of the less transparent market, this strategy may involve price manipulation. Informed dealers may refrain from trading in the more transparent market in order to exploit their informational advantage in the less transparent one. Alternatively, they may use the more transparent market in order to manipulate prices. That is, they can take advantage of the fact that the other dealers react to their trading in the more transparent market in order to shape their expectations about the value of the asset he has superior information about.

We showed that both strategies (hiding and price manipulation) increase market depth in the more transparent market.

We brought these conclusions to the data by estimating dealers strategies in the Italian Treasury bond market, by jointly considering *both* the primary and the secondary markets. In particular, we showed that informed dealers resort to the interdealer market to place sell orders at the time they have a higher informational advantage. At the same time they aggressively place bids in the primary market and buy back when the primary market closes. This strategy generates losses in the more transparent market (secondary market) for the period when the less transparent market (primary market) is open and then produces gains once the possibility of affecting the primary market is over.

These results shed new light on the old problem of price manipulation, showing how it may increase market depth. As a future research it would be interesting to analyze how the behavior of dealers in a standardized market such as the one we analyzed differs in the case we considered a less anonymous way of transacting. Also, the fact that the market of Treasury bonds is mostly affected by public news, and that idiosyncratic private information plays a minor role offers a new set of interesting problems, completely different from those traditionally found in the equity market.

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8 Appendix

Conjectured Equilibrium

We conjecture the existence of a linear equilibrium price system for the primary market and the secondary markets with price functions of the form:

$$P_a = \frac{1}{\beta^l} [X_a^i + X_a^u - 1], \quad (24)$$

$$P_1 = \mu_1 + \lambda_1 Y_1, \quad (25)$$

$$P_2 = \mu_2 + \lambda_2 Y_2. \quad (26)$$

and demand functions for the informed ("i") trader and the speculative uninformed dealer ("u") equal to:

$$X_1^i = \alpha_1 + \beta_1 V, \quad (27)$$

$$X_2^i = \alpha_2 + \beta_2 V, \quad (28)$$

$$X_a^i = \alpha_a^i + \beta_a^i [EV|P_1] + \gamma_a^i V, \quad (29)$$

$$X_a^u = \alpha_a^u + \beta_a^u [EV|P_1], \quad (30)$$

where the subscripts "1", "2" and "a" refer, respectively, to the secondary market first period, secondary market second period and auction market.

Market Maker Inference

We first focus on the inference problem of the market maker in the secondary market. In period 1 he has a value prior of $V \sim N(V_0, \Sigma_0)$, which he can update given he observes order flow $Y_1 = X_1 + u_1 = \alpha_1 + \beta_1 V + u_1$. Uninformed demand in the secondary market (u_1) is normally distributed with zero mean and variance equal to Σ_u^2 . That is, $u_t \sim N(0, \Sigma_u^2)$, for $t = 1, 2$.

The market maker observes the order flow (Y_1) and sets a price:

$$P_1 = E[V|Y_1] = V_0 + \frac{\beta_1 \Sigma_0}{\Sigma_u^2 + \beta_1^2 \Sigma_0} (Y_1 - \alpha_1 - \beta_1 V_0) = \mu_1 + \lambda_1 Y_1. \quad (31)$$

where $\mu_1 = V_0 - \frac{\beta_1 \Sigma_0}{\Sigma_u^2 + \beta_1^2 \Sigma_0} (\alpha_1 + \beta_1 V_0)$ and $\lambda_1 = \frac{\beta_1 \Sigma_0}{\Sigma_u^2 + \beta_1^2 \Sigma_0}$.

In period 2 the market maker uses the new observed order flow to update his beliefs about the value of the asset and sets a new price quote P_2 . The new order flow is: $Y_2 = X_2 + u_2 = \alpha_2 + \beta_2 V + u_2$. We assume that uninformed demand in the secondary market (u_2) is normally distributed with zero mean and variance equal to Σ_u^2 , analogously to the first period. That is, $u_t \sim N(0, \Sigma_u^2)$, for $t = 1, 2$. Therefore the market maker sets a price:

$$P_2 = E[V|Y_1, Y_2] = P_1 + \frac{\beta_2 \Sigma_1}{\Sigma_u^2 + \beta_2^2 \Sigma_1} (Y_2 - \alpha_2 - \beta_2 P_1) = \mu_2 + \lambda_2 Y_2. \quad (32)$$

where $\mu_2 = P_1 - \frac{\beta_2 \Sigma_1}{\Sigma_u^2 + \beta_2^2 \Sigma_1}(\alpha_2 + \beta_2 P_1)$ and $\lambda_2 = \frac{\beta_2 \Sigma_1}{\Sigma_u^2 + \beta_2^2 \Sigma_1}$. The term $\Sigma_1 = \frac{\Sigma_u^2 \Sigma_0}{\Sigma_u^2 + \beta_1^2 \Sigma_0}$ is the variance of the posterior in the first period.

Second Period Equilibrium in the Secondary Market

In order to solve for the equilibrium we use the standard approach (Kyle 1985) extended to a two-period set-up. We start in period 2 and we solve the optimization for the informed trader. The informed trader determines his optimal demand X_2^i which maximizes the period 2 profits Π_2^i . Therefore, in the second period, the informed dealers solve the following problem:

$$\text{Max}_{X_2^i} E[V - P_2(X_2 + u_2)] X_2 | V, P_1. \quad (33)$$

Substituting for the conjectured linear demand function ($X_2^i = \alpha_2 + \beta_2 V$) and linear price function ($P_2 = \mu_2 + \lambda_2 Y_2$) and solving for X_2^i , we get the optimal demand of bond of the informed dealers in the secondary period. It is:

$$X_2 = \frac{V - \mu_2}{2\lambda_2}. \quad (34)$$

We then impose zero profit condition and let market maker set the price equal to the expected value of the asset ($P_2 = E[V|Y_2]$), conditional upon the information he derives from the order flow ($Y_2 = X_2 + u_2$). Substitution into the price function implies parameters $\alpha_2 = \frac{-\mu_2}{2\lambda_2} = \frac{-P_1}{2\lambda_2} = -P_1\beta_2$, $\lambda_2 = \frac{1}{2} \frac{\Sigma_1^{\frac{1}{2}}}{\Sigma_u}$ and $\beta_2 = \frac{1}{2\lambda_2} = \frac{\Sigma_u}{\Sigma_1^{\frac{1}{2}}}$. Also, given that $\Sigma_1 = \frac{\Sigma_0 \Sigma_u}{\Sigma_u + \beta_1^2 \Sigma_0}$, we can rewrite:

$$\lambda_2 = \frac{\Sigma_0^{\frac{1}{2}}}{2(\Sigma_u + \beta_1^2 \Sigma_0)^{\frac{1}{2}}} \text{ and } \beta_2 = \frac{(\Sigma_u + \beta_1^2 \Sigma_0)^{\frac{1}{2}}}{\Sigma_0^{\frac{1}{2}}}. \quad (35)$$

Uninformed Speculative Demand at the Auction

The non informed speculative traders submit a demand schedule that is a function of the difference between the actual price in the market and the expected value of the asset they infer by observing prices. In particular, they maximize:

$$\text{Max}_{x_1} \{E^u[V - P_a] X_a^u | P_1\} \quad (36)$$

Using the conjectured functional forms for the bidding function of the informed traders as well as of the uninformed dealers and for the price at the auction, we find that:

$$X_a^u = -\alpha^i - 1 + (\beta^l - \beta^i)E[V|P_1]. \quad (37)$$

Equilibrium at the Auction

In period 1 the informed trader simultaneously determines the primary and secondary demand X_1^i , and X_a^i so as to maximize total profits $\Pi_a^i + \Pi_1^i + \Pi_2^i$. In particular, he solves:

$$\begin{aligned}
Max_{x_1, x_a^i} \{\Pi\} &= Max_{x_1, x_a^i} \{\Pi_a^i + \Pi_1^i + \Pi_2^i\} = \\
&= Max_{x_1, x_a^i} \{E[V - P_1] X_1 + E[V - P_a] X_a^i + \Pi_2(P_1)\} = \\
&= Max_{x_1, x_a^i} \{E[V - \mu_1 - \lambda_1(Y_1 + u_1)] X_1 + E\left[V - \frac{1}{\beta^l} (X_a^i + X_a^u - 1)\right] X_a^i + \\
&\quad \frac{1}{4\lambda_2} [(V - \mu_1 - \lambda_1 X_1)^2 + \lambda_1^2 \Sigma_u^2]\} \tag{38}
\end{aligned}$$

where the conjectured functional forms for the demand in the secondary market, the bidding schedules at the auction and the price function in the primary and secondary markets have been used. $Y_a = X_a^i + X_a^u + X_a^l - 1$ and $Y_1 = X_1 + u_1$.

Solving equation 38 with respect to X_a^i , we find that the informed dealers' demand at the auction is:

$$X_a^i = 1 + \frac{1}{6} \beta^l E^u [V|P_1] + \frac{1}{2} \beta^l V. \tag{39}$$

By using the conjectural forms of the bidding schedule, we find that the bidding schedule of the uninformed speculative dealers is:

$$X_a^u = \frac{5}{6} \beta^l E^u [V|P_1]. \tag{40}$$

Also, the price function at the action is:

$$P_a = \frac{1}{\beta^l} (X_a^i + X_a^u - 1) = \lambda_a (X_a^i + X_a^u - 1) = \frac{1}{2} V + E[V^u|P_1]. \tag{41}$$

Given that we have assumed that $E[V^u|P_1] = \theta_\beta P_1$, we have:

$$X_a^i = 1 + \frac{1}{6} \beta^l \theta_\beta P_1 + \frac{1}{2} \beta^l V, \quad X_a^u = \frac{5}{6} \beta^l \theta_\beta P_1 \quad \text{and} \quad P_a = \frac{1}{2} V + \theta_\beta P_1 \tag{42}$$

Therefore, the equilibrium at the auction is fully specified in terms of the exogenous parameters: β^l and θ_β .

First Period Equilibrium in the Secondary Market

Solving equation 38 with respect to X_1 , we find that:

$$X_1 = \alpha_1 + \beta_1 V, \tag{43}$$

where $\alpha_1 = -\frac{36\mu_1\lambda_2 + \lambda_1[6\theta_\beta\lambda_2 + \mu_1(-18 + \beta^l\theta_\beta^2\lambda_2)] + \beta^l\lambda_1^2\lambda_2\theta_\beta^2u}{\lambda_1[-18\lambda_1 + 72\lambda_2 + \beta^l\lambda_1\lambda_2\theta_\beta]}$, $\beta_1 = \frac{3[12\lambda_2 - \lambda_1(6 + \beta^l\theta_\beta\lambda_2)]}{\lambda_1[-18\lambda_1 + 72\lambda_2 + \beta^l\lambda_1\lambda_2\theta_\beta]}$.

From equations 35 we know that $\lambda_2 = \frac{\Sigma_0^{\frac{1}{2}}}{2(\Sigma_u + \beta_1^2 \Sigma_0)^{\frac{1}{2}}}$ and from 32 we know that $\lambda_1 = \frac{\beta_1 \Sigma_0}{\Sigma_u^2 + \beta_1^2 \Sigma_0}$. Combining the two equations and using the focs, (equation 43), we get:

$$\frac{\lambda_1}{4\lambda_2^2} - \frac{3[12\lambda_2 - \lambda_1(6 + \beta^l\theta_\beta\lambda_2)]}{\lambda_1[-18\lambda_1 + 72\lambda_2 + \beta^l\lambda_1\lambda_2\theta_\beta^2]} = 0. \tag{44}$$

In order to solve the model we make the simplifying assumption that $\beta^l = \frac{1}{\lambda_1 \theta_\lambda}$. The rationale is the following. We know from the market clearing condition that: $P_a = \frac{1}{\beta^l} (X_a^i + X_a^u - 1)$. We can therefore interpret $\frac{1}{\beta^l}$ as the equivalent of the market depth in the primary market, that is the impact of the submitted bids on the price schedule. We define θ_λ as the parameter that relates the market depth in the primary and secondary market (i.e., $\frac{1}{\beta^l} = \theta_\lambda \lambda_1$). The model is now parametric in terms of two parameters: θ_λ and θ_β . Therefore, equation 44 becomes:

$$\frac{\lambda_1}{4\lambda_2^2} + \frac{3[12\lambda_2 + \lambda_1(6 + \beta^l \lambda_2 \theta_\beta)]}{\lambda_1[-18\lambda_1 + 72\lambda_2 + \beta^l \lambda_1 \lambda_2 \theta_\beta^2]} = 0. \quad (45)$$

Substituting for $w = \frac{\lambda_2}{\lambda_1}$, we get the cubic equation:

$$12(\theta_\beta - 12\theta_\lambda)w^3 + 72\theta_\lambda w^2 + (\theta_\beta^2 + 72\theta_\lambda)w - 18\theta_\lambda = 0. \quad (46)$$

There are three sets of solution for each pair of θ_λ and θ_β values. We discard all the solutions which provide imaginary numbers and then we restrict the sets of solutions to the ones that satisfy the following restrictions:

$$P_1 > 0, P_2 > 0, P_a > 0, X_a^i > 0, X_a^u > 0, \lambda_1 > 0 \text{ and } \lambda_2 > 0. \quad (47)$$

This yields the optimal values:

$$\beta_1 = \frac{1}{\Gamma} \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}, \lambda_1 = \frac{\Gamma}{(1 + \Gamma^2)} \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u}, \text{ and } \Sigma_1 = \Sigma_0 \frac{1}{1 + \Gamma^2}. \quad (48)$$

where $\Gamma = \sqrt{4w^2 - 1}$. The solutions for different sets of values of θ_λ and θ_β are provided in the Figures 1-3. In particular, we consider the cases where $0 < \theta_\lambda < 10$ and $0 < \theta_\beta < 10$. The value of μ_1 can be analogously derived.

Corollary 1

The equilibrium values for the no-auction days are derived by substituting in equations ?? and 48 the values $\theta_\lambda = 1$ and $\theta_\beta = 0$.

$$\beta_1^{NA} = 0.6671 \frac{\Sigma_u}{\Sigma_0^{\frac{1}{2}}}, \beta_2^{NA} = 1.2021 \frac{\Sigma_u}{\Sigma_1^{\frac{1}{2}}} = \frac{\sqrt{2}\Sigma_u}{\Sigma_0^{\frac{1}{2}}}, \lambda_1^{NA} = 0.4617 \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u} \quad (49)$$

$$\lambda_2^{NA} = \frac{1}{2} \frac{\Sigma_1^{\frac{1}{2}}}{\Sigma_u} = 0.4159 \frac{\Sigma_0^{\frac{1}{2}}}{\Sigma_u} \text{ and } \Sigma_1^{NA} = \Sigma_0 \frac{1}{1 + \beta_1^2 \frac{\Sigma_0}{\Sigma_u}} = 0.6920 \Sigma_0. \quad (50)$$

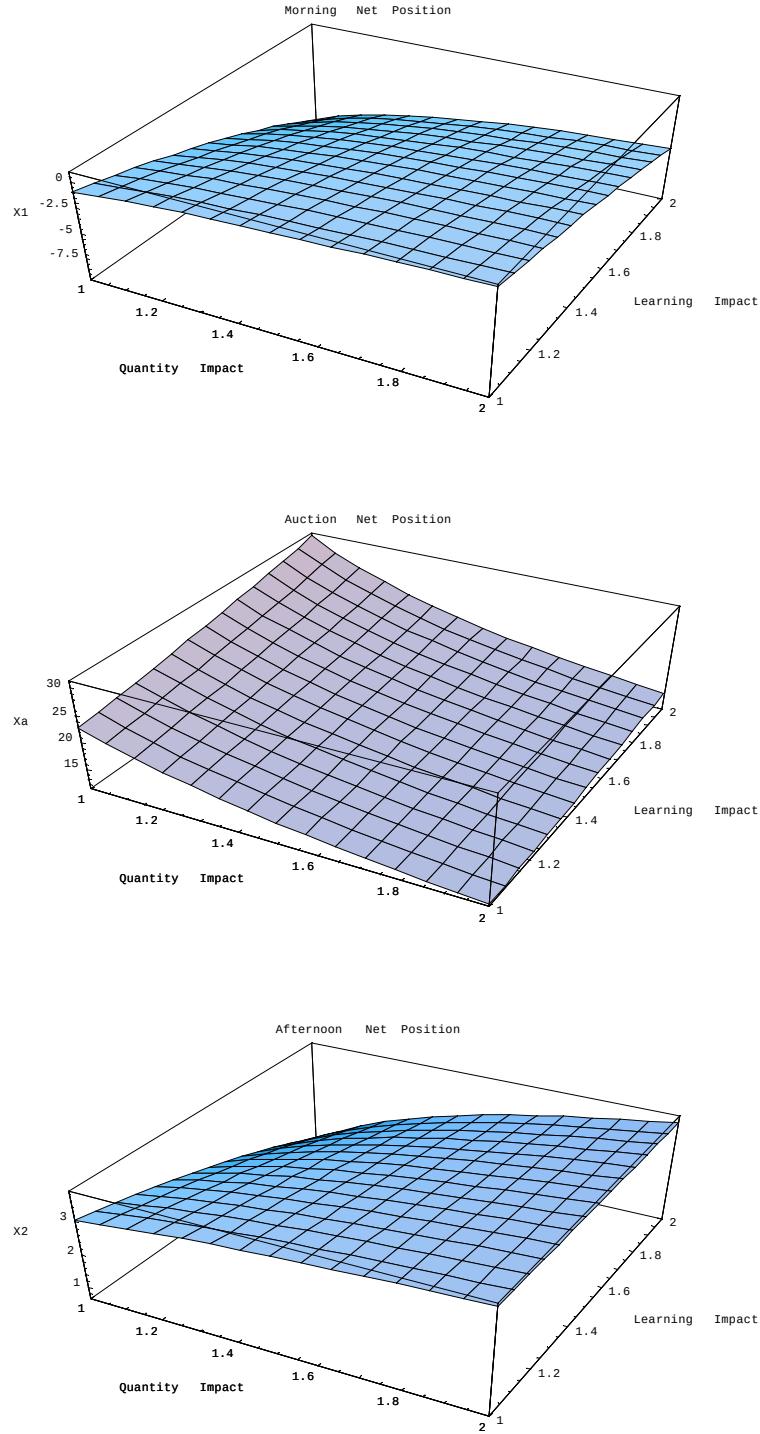


Figure 1: Informed positions in the primary market (X_{49}) and in the secondary market in the morning (X_1) and in the afternoon (X_2), for different values of learning impact (θ_β) and quantity impact (θ_λ). The values of $\Sigma_o = 0.5$ and $\Sigma_u = 0.5$.

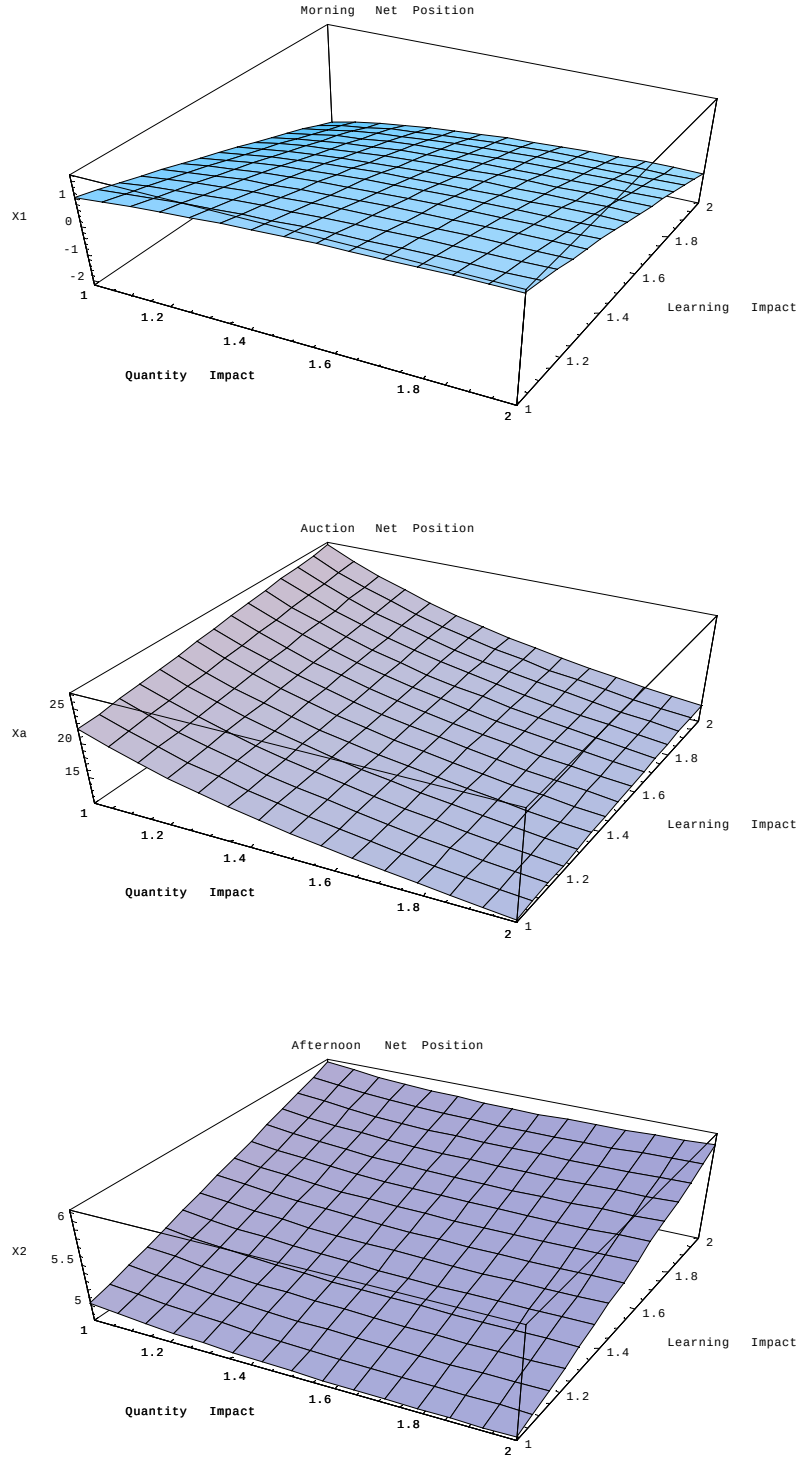


Figure 2: Informed positions in the primary market (X_a^{50}) and in the secondary market in the morning (X_1) and in the afternoon (X_2) for different values of learning impact (θ_β) and quantity impact (θ_λ). The values of $\Sigma_o = 5$ and $\Sigma_u = 5$.

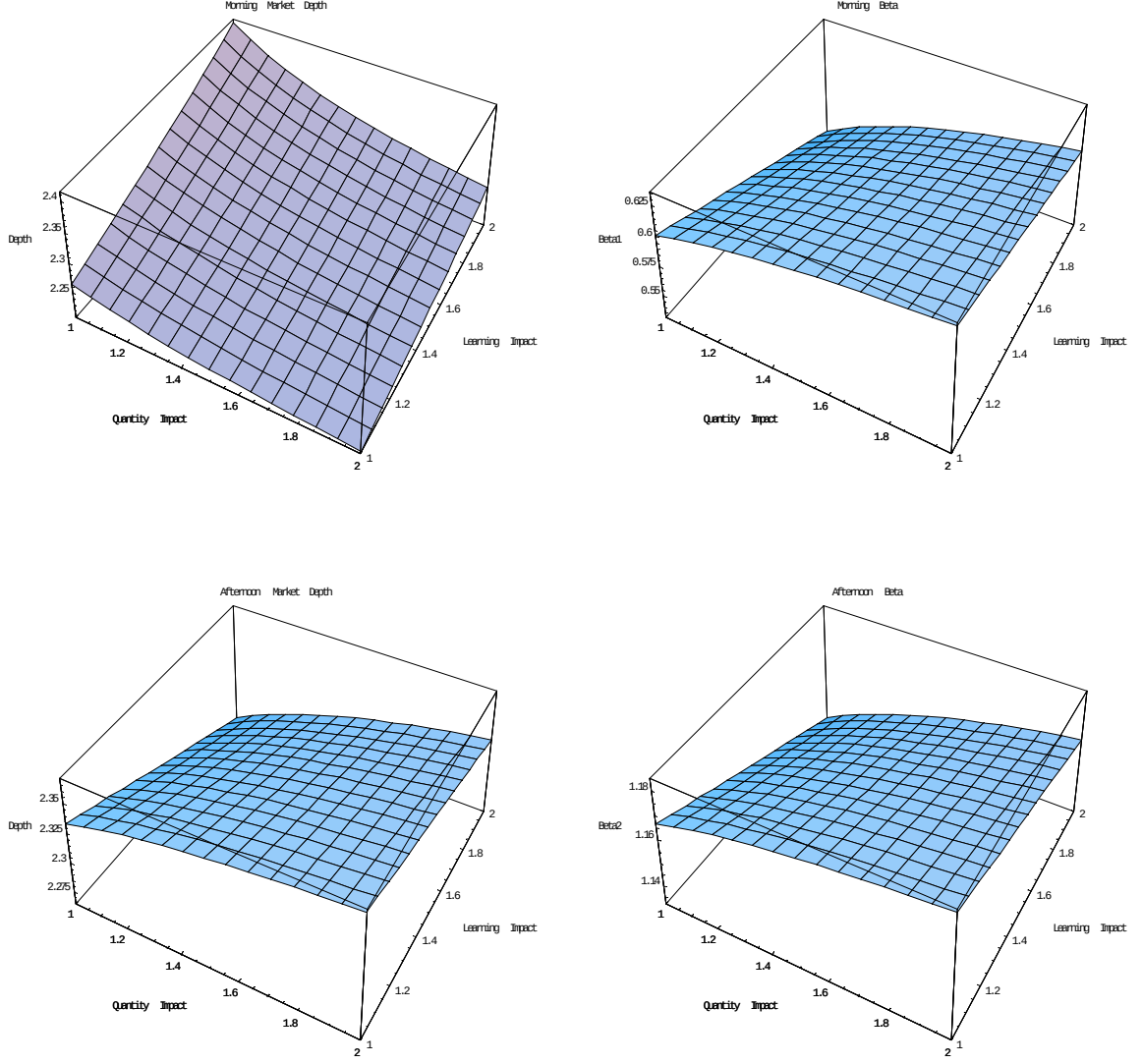


Figure 3: Values of β_1 , β_2 , λ_1 and λ_2 for different values of learning impact (θ_β) and quantity impact (θ_λ). The subscript "1" and "2" refer, respectively, to the morning and the afternoon. The values of $\Sigma_o = 0.5$ and $\Sigma_u = 0.5$.

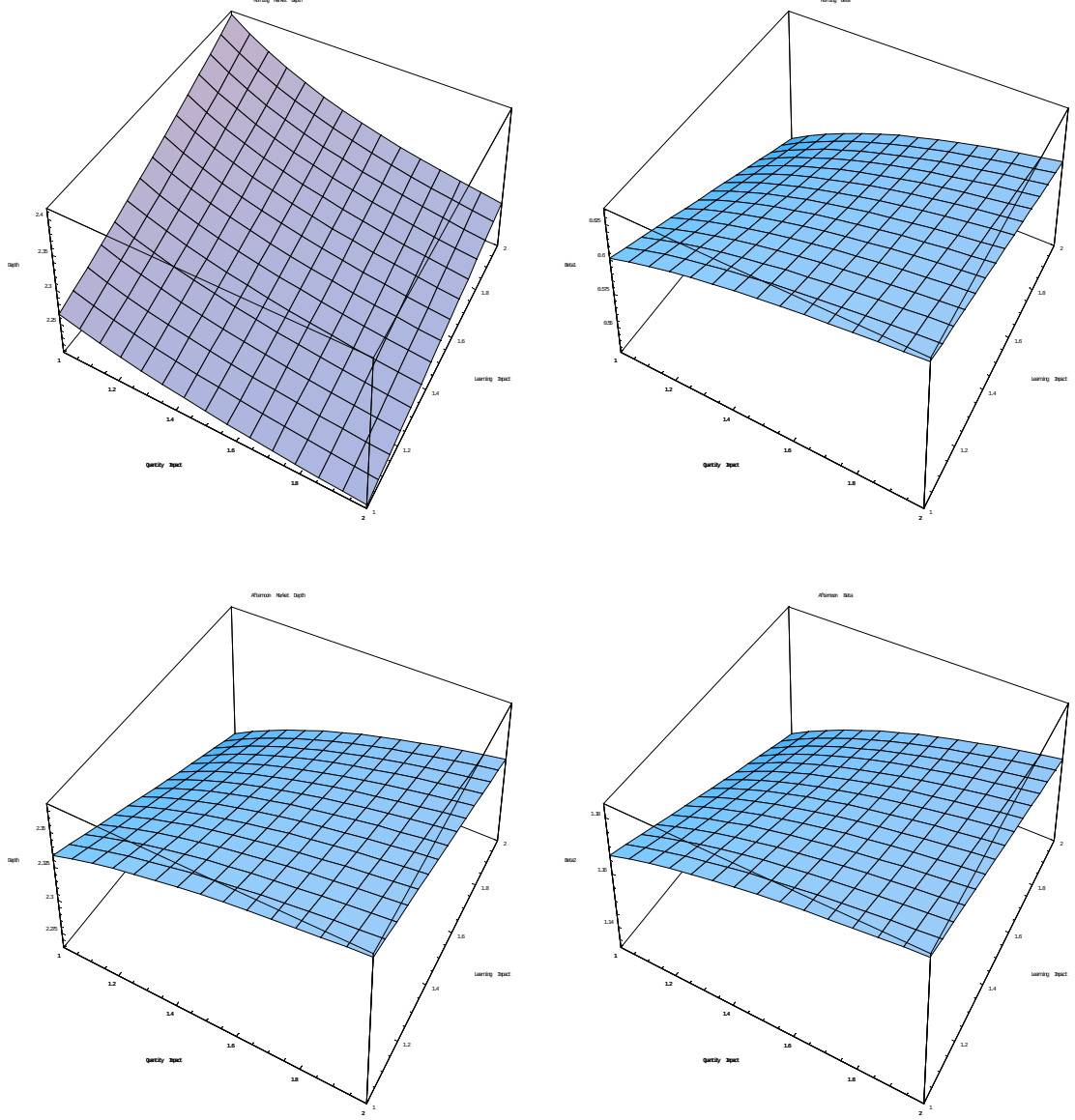


Figure 4: values of β_1 , β_2 , λ_1 and λ_2 for different values of learning impact (θ_β) and quantity impact (θ_λ). The subscript "1" and "2" refer, respectively, to the morning and the afternoon. The values of $\Sigma_o = 5$ and $\Sigma_u = 5$.

TABLE 1 Transactions on the secondary market

The number of transactions and volume are reported. Volume is defined as number of transactions multiplied by the their nominal values expressed in billions of liras. The minimum unitary size of each transaction is 5 million liras. Each transaction corresponds to a multiple of such a value. The transactions have been broken down according to the type of bond involved and according to the type of dealer who *originated* them.

(breakdown by type of bond)			(breakdown by type of dealer)		
Type of Bond	Transaction Number	Mean Volume	Type of Dealer	Transaction Number	Mean Volume
T. Bonds	1,081,945	17,990	Specialists	505,594	8,660
F.I. Bonds	301,306	5,590	Ordinary Market Makers	536,559	9,020
T. Note	10,186	190	Ordinary Dealers	351,284	6,090
TOTAL	1,393,437	23,770	TOTAL	1,393,437	23,770

TABLE 2 Dispersion of bids in the primary market**Panel A**

TB represents the total number of bids submitted, WB is the total number of winning bids, SD is their dispersion, measured as standard deviations and NSB are the non-speculative bids. The latter represent the total demand, net of those bids based on a price which differed for more than 2 liras from the actual auction prices and not truncated by the truncation mechanism devised to eliminate the demands too deviant. The sample covers all the auctions on the fixed income bonds (T. Bonds), financially indexed bonds (F.I. Bonds) and the 2-year zero-coupon bonds (T. Notes), in the period February 94-February 96.

Type of Bond	TB/WB	SB/WB	SD/TB	NSB/WB
T. Bonds	1.81	120	125	1.50
F.I. Bonds	2.31	360	385	2.08
T. Note	2.01	210	217	1.95
TOTAL	2.05	237	242	1.85

Panel B

Herfindhal and Entropy are indexes of concentration. G10 and G20 Indexes represent the share of total bids (submitted or winning) of, respectively, the 10 and 20 biggest dealers over total. The sample covers all the auctions on the fixed income bonds (T. Bonds), financially indexed bonds (F.I.Bonds) and the 2-year zero-coupon bonds (T. Notes), in the period February 94-February 96.

Type of Bond	Herfindhal Ind		Entropy Ind.		C10 Index		C20 Index	
	Total Bids	Winning Bids	Total Bids	Winning Bids	Total Bids	Winning Bids	Total Bids	Winning Bids
T. Bonds	0.061	0.092	-3.147	-2.863	64.81	74.81	88.00	93.61
F.I. Bonds	0.051	0.072	-3.284	-2.991	59.75	70.96	84.26	92.58
T. Note	0.080	0.095	-3.064	-2.912	66.61	71.58	88.11	92.53
TOTAL	0.064	0.087	-3.165	-2.922	63.72	72.45	86.79	92.91

TABLE 3 Trades and Profits with and without parallel markets

We report the average values of profits and trades for the different classes of dealers for the day of the auction ($Trade_{auct}$ and $Profit_{auct}$) and for the days before the auction ($Trade_{noauct}$ and $Profit_{noauct}$) and the statistics (p -value and t -stat) of the t -tests of the difference between the two. Dealers are grouped in 5 classes on the basis of the profits they make at the auction. These profits are equal to the difference between the cut-off price at the auction and the price at the end of the day multiplied by the number of bonds they get allotted at the auction. Dealers are ranked according to these profits and grouped 5 categories: the category made of the first 10 dealers (G10), the category made of the first 15 dealers (G15), the category made of the first 20 dealers (G20), the category made of the first 30 dealers (G30) and the category made of the dealers ranking between 50 and 100 (Others). The profits in the secondary market are equal to the quantities purchased multiplied by the difference between the price at the end of the day and the price at which the transaction is executed in the case of purchases and to the quantities sold multiplied by the difference between the price at which the transaction is executed and the price at the end of the day in the case of sales. Trades are the equal to the difference between purchases and sales. Received trades are the transactions the dealers carry out to execute incoming orders, while originated trades are the transactions originated by orders the dealer directly places with other dealers. Morning refers to the period when both the primary market and secondary market are open (9 a.m.-1 p.m). Afternoon refers to the period when after the auction deadline and before the announcement of the results (1 p.m.-announcement time). In the afternoon, the primary market is closed and the secondary market is open.

Panel A: Morning

Group	Received Trade				Originated Trade			
	$Trade_{auct}$	$Trade_{noauct}$	$Ttrade_{auct} - Trade_{noauct}$		$Trade_{auct}$	$Trade_{noauct}$	$Ttrade_{auct} - Trade_{noauct}$	
	Mean	Mean	P-value	T-stat	Mean	Mean	P-value	T-stat
G10	-0.51	-4.61	0.00	7.20	-3.40	2.16	0.00	-7.94
G15	-0.40	-4.36	0.00	8.97	-2.63	2.37	0.00	-9.34
G20	-0.06	-3.94	0.00	10.78	-1.91	2.56	0.00	-10.44
G30	0.10	-3.14	0.00	12.44	-1.11	2.36	0.00	-11.35
Others	0.09	-0.15	0.01	2.40	0.31	0.21	0.41	0.81

Group	Profits form Received Trade				Profits from Originated Trade			
	$Profit_{auct}$	$Profit_{noauct}$	$Profit_{auct} - Profit_{noauct}$		$Profit_{auct}$	$Profit_{noauct}$	$Profit_{auct} - Profit_{noauct}$	
	Mean	Mean	P-value	T-stat	Mean	Mean	P-value	T-stat
G10	0.0006	-0.7446	0.0000	5.6827	0.1543	0.8820	0.0001	-3.92
G15	0.0202	-0.7286	0.0000	6.8279	0.0781	0.7814	0.0000	-4.90
G20	0.0020	-0.6614	0.0000	7.3660	0.0737	0.6927	0.0000	-5.55
G30	-0.0111	-0.5251	0.0000	7.9310	0.0330	0.5316	0.0000	-6.25
Others	0.0215	0.0005	0.1745	1.3580	0.1543	0.0402	0.0024	-3.03

Panel B: Afternoon

Group	Received Trade				Originated Trade			
	$Trade_{auct}$	$Trade_{noauct}$	$Ttrade_{auct} - Trade_{noauct}$		$Trade_{auct}$	$Trade_{noauct}$	$Ttrade_{auct} - Trade_{noauct}$	
	Mean	Mean	P-value	T-stat	Mean	Mean	P-value	T-stat
G10	-0.44	-4.61	0.00	14.90	0.35	2.16	0.32	-6.66
G15	-0.41	-4.36	0.00	17.66	0.35	2.37	0.00	-9.43
G20	-0.29	-3.94	0.00	19.70	0.31	2.56	0.00	-12.55
G30	-0.26	-3.14	0.00	21.02	0.23	2.36	0.00	-15.77
Others	-0.0009	-0.15	0.94	7.76	-0.04	0.21	0.01	-7.12

Group	Profits form Received Trade				Profits from Originated Trade			
	$Profit_{auct}$	$Profit_{noauct}$	$Profit_{auct} - Profit_{noauct}$		$Profit_{auct}$	$Profit_{noauct}$	$Profit_{auct} - Profit_{noauct}$	
	Mean	Mean	P-value	T-stat	Mean	Mean	P-value	T-stat
G10	-0.05	-0.74	0.00	13.13	0.11	0.88	0.00	-12.41
G15	-0.05	-0.72	0.00	15.59	0.08	0.78	0.00	-15.17
G20	-0.05	-0.66	0.00	17.24	0.06	0.69	0.00	-17.01
G30	-0.03	-0.52	0.00	19.33	0.04	0.53	0.00	-18.20
Others	0.001	0.0005	0.87	0.16	0.0005	0.04	0.15	-5.24

TABLE 4
Dealers' strategies: Trading positions.

We estimate $\Delta P_t = a + bB_{j,t} + gS_{j,t} + dC_t + e_t$, where ΔP_t is, alternatively, $(P_{t+1,A} - P_{t,B})$ or $(P_{t+1,A} - P_{t,A})$, $B_{j,t}$ and $S_{j,t}$ are the purchases and sales of the j th dealer and C_t is a vector of control variables. They include the duration of the bond being auctioned off (*Duration*), the Herfindhal index of concentration of the bids in the primary market (*Herfindhal*), the cover ratio - i.e., the ratio between total demand and total amount of bonds allotted in the primary market - (*Dom/Allot*), the dispersion of both the submitted bids and the winning bids in the primary market (respectively *Volat Demand* and *Volat Allott.*) and the volatility in the secondary market (*Volat. Sec. Mkt*). Dealers are grouped in 5 classes on the basis of the profits they make at the auction. These profits are equal to the difference between the cut-off price at the auction and the price at the end of the day multiplied by the number of bonds they get allotted at the auction. Dealers are ranked according to these profits and grouped 5 categories: the category made of the first 10 dealers (G10), the category made of the first 15 dealers (G15), the category made of the first 20 dealers (G20), the category made of the first 30 dealers (G30) and the category made of the dealers ranking between 50 and 100 (Others). Received trades are the transactions the dealers carry out to execute incoming orders, while originated trades are the transactions originated by orders the dealer directly places with other dealers. Morning refers to the period when both the primary market and secondary market are open (9 a.m.-1 p.m). Afternoon refers to the period when after the auction deadline and before the announcement of the results (1 p.m.-announcement time). In the afternoon, the primary market is closed and the secondary market is open. The model is estimated through a pooled regression, with a White's robust variance-covariance matrix.

Panel A: Trades originated by the dealers in the morning

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-3.61	-1.05	-4.02	-1.48	-3.84	-1.63	-3.85	-2.04	-4.19	-2.74	-5.42	-1.57	-5.83	-2.14	-5.69	-2.41	-5.78	-3.05	-6.24	-4.07
Buys	-0.005	-0.27	-0.02	-1.52	-0.01	-1.23	-0.01	-1.45	-0.01	-0.96	0.003	0.17	-0.01	-0.93	-0.009	-0.64	-0.01	-0.87	-0.005	-0.34
Sells	0.03	2.21	0.05	2.90	0.05	3.16	0.05	3.19	0.0009	0.11	0.03	2.12	0.05	2.86	0.05	3.11	0.05	3.18	-0.003	-0.46
Herfindhal	24.51	1.66	25.37	2.11	24.88	2.40	24.83	2.95	24.56	3.43	-9.23	-0.59	-8.17	-0.64	-8.50	-0.78	-8.43	-0.95	-8.45	-1.12
Dom/Ass	4.28	3.24	4.37	4.13	4.34	4.79	4.38	6.02	4.52	7.66	4.19	3.10	4.27	3.94	4.26	4.59	4.32	5.80	4.51	7.46
Volat Demand	-0.02	-0.85	-0.02	-1.13	-0.02	-1.29	-0.02	-1.68	-0.03	-2.32	0.01	0.57	0.01	0.65	0.01	0.77	0.01	0.85	0.009	0.71
Volat Assign	0.10	2.82	0.10	3.59	0.10	4.14	0.10	5.27	0.11	6.75	0.06	1.71	0.06	2.21	0.06	2.56	0.06	3.34	0.07	4.46
Volat Sec. Mkt	-0.51	-5.56	-0.50	-6.78	-0.50	-7.82	-0.50	-9.63	-0.49	-11.44	-0.40	-4.80	-0.39	-5.84	-0.39	-6.75	-0.39	-8.28	-0.37	-9.77
Duration	-0.43	-2.42	-0.43	-2.99	-0.44	-3.52	-0.43	-4.24	-0.40	-4.30	-0.17	-0.98	-0.17	-1.21	-0.18	-1.46	-0.17	-1.73	-0.13	-1.48
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
ARsquare	0.08		0.08		0.08		0.08		0.08		0.09		0.09		0.09		0.09		0.08	

Panel B: Trades originated by the dealers in the afternoon

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-4.63	-1.57	-4.04	-1.61	-4.35	-2.01	-4.28	-2.41	-4.20	-2.74	-6.72	-2.27	-6.14	-2.45	-6.43	-2.98	-6.35	-3.57	-6.26	-4.07
Buys	0.42	2.98	0.46	3.43	0.45	3.57	0.40	3.73	0.23	1.53	0.41	2.92	0.46	3.40	0.45	3.52	0.40	3.72	0.23	1.56
Sells	-0.14	-2.04	-0.11	-1.36	-0.11	-1.57	-0.13	-2.06	-0.02	-0.23	-0.17	-2.33	-0.13	-1.59	-0.13	-1.75	-0.14	-2.17	0.008	0.06
Herfindhal	21.50	1.44	19.29	1.56	22.43	2.11	22.79	2.66	24.64	3.44	-11.22	-0.71	-13.58	-1.05	-10.46	-0.93	-10.19	-1.13	-8.41	-1.11
Dom/Ass	4.19	3.52	4.06	4.07	4.20	4.89	4.32	6.20	4.52	7.64	4.20	3.44	4.06	3.98	4.20	4.78	4.31	6.05	4.51	7.45
Volat Demand	-0.02	-0.90	-0.02	-0.83	-0.02	-1.31	-0.02	-1.73	-0.03	-2.33	0.01	0.55	0.02	0.89	0.01	0.72	0.01	0.80	0.009	0.70
Volat Assign	0.10	2.91	0.09	3.12	0.10	4.02	0.10	5.20	0.11	6.76	0.06	1.82	0.05	1.86	0.06	2.53	0.06	3.32	0.07	4.47
Volat Sec. Mkt	-0.42	-4.80	-0.43	-5.97	-0.44	-7.06	-0.45	-9.07	-0.49	-11.40	-0.30	-3.82	-0.31	-4.73	-0.32	-5.68	-0.33	-7.47	-0.37	-9.72
Duration	-0.38	-2.10	-0.40	-2.76	-0.40	-3.13	-0.41	-3.95	-0.40	-4.36	-0.11	-0.65	-0.13	-0.95	-0.13	-1.07	-0.14	-1.44	-0.14	-1.55
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
ARsquare	0.13		0.14		0.12		0.11		0.07		0.13		0.14		0.12		0.11		0.08	

Panel C: Trades received by the dealers in the morning

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-2.94	-0.83	-2.98	-1.06	-3.21	-1.33	-3.61	-1.89	-4.18	-2.73	-4.80	-1.35	-4.85	-1.72	-5.07	-2.10	-5.52	-2.88	-6.21	-4.04
Buys	0.03	1.10	0.06	2.22	0.06	2.43	0.06	2.70	-0.002	-0.33	0.02	1.03	0.06	2.21	0.06	2.44	0.06	2.79	0.0006	0.08
Sells	0.01	0.99	-0.007	-0.39	-0.008	-0.49	-0.01	-1.08	-0.01	-0.85	0.02	1.41	-0.001	-0.07	-0.001	-0.08	-0.01	-0.73	-0.02	-1.47
Herfindhal	22.39	1.52	22.33	1.85	22.39	2.15	23.26	2.75	24.51	3.42	-11.06	-0.71	-11.10	-0.87	-11.08	-1.01	-10.09	-1.13	-8.64	-1.14
Dom/Ass	4.08	3.04	4.04	3.732	4.15	4.50	4.29	5.84	4.51	7.64	4.006	2.91	3.96	3.58	4.06	4.31	4.23	5.62	4.50	7.44
Volat Demand	-0.02	-0.69	-0.01	-0.78	0.02	-1.01	-0.02	-1.47	-0.03	-2.31	0.02	0.70	0.02	0.94	0.02	1.00	0.01	1.03	0.01	0.74
Volat Assign	0.10	2.59	0.09	3.15	0.10	3.76	0.10	4.97	0.11	6.73	0.05	1.52	0.05	1.84	0.05	2.22	0.06	3.06	0.07	4.43
Volat Sec. Mkt	-0.52	-5.45	-0.52	-6.74	-0.51	-7.83	0.50	-9.64	-0.49	-11.43	-0.40	-4.67	-0.40	-5.75	-0.40	-6.70	-0.39	-8.25	-0.37	-9.78
Duration	-0.46	-2.75	-0.47	-3.42	-0.45	-3.75	-0.44	-4.33	-0.40	-4.32	-0.20	-1.25	-0.21	-1.57	-0.19	-1.66	-0.18	-1.81	-0.13	-1.48
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.08		0.09		0.09		0.08		0.08		0.09		0.10		0.09		0.09		0.08	

Panel D: Trades received by the dealers in the afternoon

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-4.17	-1.35	-3.66	-1.37	-3.82	-1.67	-3.93	-2.12	-0.62	-1.41	-6.22	-2.00	-5.72	-2.14	-5.89	-2.56	-5.97	-3.22	-6.36	-4.13
Buys	0.11	0.40	-0.06	-0.33	-0.10	-0.64	-0.09	-0.61	0.01	0.20	0.11	0.37	-0.07	-0.34	-0.11	-0.67	-0.09	-0.61	-0.22	-1.76
Sells	0.44	2.09	0.47	2.87	0.53	3.39	0.50	3.66	0.10	1.00	0.43	1.95	0.47	2.77	0.53	3.28	0.51	3.55	0.82	2.59
Herfindhal	15.74	1.05	19.13	1.54	19.67	1.84	21.01	2.43	-43.31	-7.28	-17.09	-1.08	-13.89	-1.06	-13.30	-1.18	-12.03	-1.32	-7.96	-1.05
Dom/Ass	4.009	3.31	3.88	3.74	3.97	4.44	4.11	5.73	0.87	7.66	4.00	3.22	3.88	3.65	3.96	4.33	4.10	5.58	4.53	7.48
Volat Demand	-0.01	-0.61	-0.01	-0.71	-0.01	-0.89	-0.02	-1.35	0.02	2.88	0.02	0.82	0.02	0.99	0.02	1.09	0.01	1.13	0.008	0.62
Volat Assign	0.09	2.67	0.09	3.01	0.09	3.57	0.10	4.83	-0.01	-1.55	0.05	1.57	0.05	1.74	0.05	2.10	0.06	2.97	0.07	4.54
Volat Sec. Mkt	-0.42	-4.56	-0.44	-5.63	-0.44	-6.56	-0.45	-8.40	0.11	5.92	-0.31	-3.55	-0.32	-4.44	-0.32	-5.18	-0.33	-6.78	-0.36	-9.66
Duration	-0.41	-2.37	-0.41	-2.83	-0.41	-3.28	-0.42	-4.06	-0.26	-10.78	-0.15	-0.87	-0.14	-1.01	-0.14	-1.19	-0.15	-1.52	-0.14	-1.54
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.14		0.12		0.12		0.11		0.08		0.14		0.13		0.13		0.12		0.08	

TABLE 5 Trading strategies in the two parallel markets.

We estimate $N_{j,t} = a + bQ_{j,t} + \beta C_t + e_t$, where $N_{j,t}$ is the net trade in the secondary market of the j th dealer, $Q_{j,t}$ is the quantity demanded at the auction by the j th dealer, alternatively. The net trade is constructed as the difference between buy and sell orders placed with other dealers. Alternatively, we consider the change in net trade between the afternoon and the morning. That is, the difference between the net trade (buy orders less sell orders) in the afternoon and the net trade in the morning. C_t is a vector of control variables. They include the duration of the bond being auctioned off (*Duration*), the Herfindhal index of concentration of the bids in the primary market (*Herfindhal*), the cover ratio - i.e., the ratio between total demand and total amount of bonds allotted in the primary market – (*Dom/Allot*), the dispersion of both the submitted bids and the winning bids in the primary market (respectively *Volat Demand* and *Volat Allot.*) and the volatility in the secondary market (*Volat. Sec. Mkt*). Dealers are grouped in 5 classes on the basis of the profits they make at the auction. These profits are equal to the difference between the cut-off price at the auction and the price at the end of the day multiplied by the number of bonds they get allotted at the auction. Dealers are ranked according to these profits and grouped 5 categories: the category made of the first 10 dealers (G10), the category made of the first 15 dealers (G15), the category made of the first 20 dealers (G20), the category made of the first 30 dealers (G30) and the category made of the dealers ranking between 50 and 100 (Others). Received trades are the transactions the dealers carry out to execute incoming orders, while originated trades are the transactions originated by orders the dealer directly places with other dealers. Morning refers to the period when both the primary market and secondary market are open (9 a.m.-1 p.m). Afternoon refers to the period when after the auction deadline and before the announcement of the results (1 p.m.-announcement time). In the afternoon, the primary market is closed and the secondary market is open. The model is estimated through a pooled regression, with a White's robust variance-covariance matrix. The values of the coefficients have been divided by 1,000.

Groups	Panel A: Net trade in the morning										Panel B: Change in trade between afternoon and morning									
	Trade originated by the dealer										Trade originated by the dealer									
	G10		G15		G20		G30		Others		G10		G15		G20		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	3.03	1.94	-16.61	-3.12	-13.32	-3.20	-7.29	-2.50	1.26	1.03	15.89	2.28	18.82	3.33	16.16	3.69	9.17	3.00	-1.49	-1.19
Demand at the auction	-0.02	-2.84	-0.02	-2.73	-0.02	-2.66	-0.01	-2.49	-0.02	-1.06	0.03	3.48	0.02	3.28	0.02	3.48	0.02	3.33	0.02	1.06
Herfindhal	68.41	1.45	51.58	1.45	33.17	1.21	21.65	1.11	4.90	0.53	-68.93	-1.48	-49.03	-1.37	-35.38	-1.29	-21.03	-1.08	-4.50	-0.48
Dom/Ass	3.86	1.86	4.54	2.78	3.66	2.89	2.19	2.50	0.08	0.22	-3.80	-1.77	-4.32	-2.51	-3.72	-2.80	-2.22	-2.43	0.001	0.002
Volat Demand	-0.08	-0.97	-0.07	-1.20	-0.05	-1.04	-0.02	-0.60	-0.008	-0.53	0.09	1.05	0.07	1.11	0.05	1.16	0.02	0.65	0.006	0.43
Volat Assign	0.16	1.55	0.14	1.81	0.11	1.82	0.06	1.47	0.01	0.77	-0.16	-1.60	-0.12	-1.63	-0.11	-1.87	-0.06	-1.44	-0.01	-0.67
Volat Sec. Mkt	-0.21	-0.83	0.13	0.65	0.13	0.79	0.04	0.34	-0.07	-2.66	0.008	0.02	-0.33	-1.52	-0.31	-1.81	-0.17	-1.44	0.07	2.61
Duration	0.11	0.28	-0.15	-0.53	-0.16	-0.69	-0.25	-1.51	-0.06	-0.77	-0.16	-0.38	0.13	0.42	0.13	0.53	0.25	1.49	0.05	0.64
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.03		0.02		0.02		0.01		0.01		0.001		0.001		0.001		0.001		0.05	

TABLE 6
Dealers' strategies: Profits.

We estimate $\Delta P_t = a + b\Pi_{b,j,t} + g\Pi_{s,j,t} + \mathbf{d}\mathbf{C}_t + e_t$, where ΔP_t is, alternatively, $(P_{t+1,A} - P_{t,B})$ or $(P_{t+1,A} - P_{t,A})$, $\Pi_{b,j,t}$ and $\Pi_{s,j,t}$ are, respectively, the profits arising out of purchases and sales of the j th dealer and $\mathbf{C}_t$ is a vector of control variables. They include the duration of the bond being auctioned off (*Duration*), the Herfindhal index of concentration of the bids in the primary market (*Herfindhal*), the cover ratio - i.e., the ratio between total demand and total amount of bonds allotted in the primary market - (*Dom/Allot*), the dispersion of both the submitted bids and the winning bids in the primary market (respectively *Volat Demand* and *Volat Allot*.) and the volatility in the secondary market (*Volat. Sec. Mkt*). Dealers are grouped in 5 classes on the basis of the profits they make at the auction. These profits are equal to the difference between the cut-off price at the auction and the price at the end of the day multiplied by the number of bonds they get allotted at the auction. Dealers are ranked according to these profits and grouped 5 categories: the category made of the first 10 dealers (G10), the category made of the first 15 dealers (G15), the category made of the first 20 dealers (G20), the category made of the first 30 dealers (G30) and the category made of the dealers ranking between 50 and 100 (Others). The profits in the secondary market are equal to the quantities purchased multiplied by the difference between the price at the end of the day and the price at which the transaction is executed in the case of purchases and to the quantities sold multiplied by the difference between the price at which the transaction is executed and the price at the end of the day in the case of sales. Received profits are the profits arising out of the transactions the dealers carry out to execute incoming orders, while originated profits are the ones arising out of the transactions originated by orders the dealer directly places with other dealers. Net Profits is the sum of the received and originated profits. Morning refers to the period when both the primary market and secondary market are open (9 a.m.-1 p.m). Afternoon refers to the period when after the auction deadline and before the announcement of the results (1 p.m.-announcement time). In the afternoon, the primary market is closed and the secondary market is open.

Panel A: Profits from trades originated by the dealers in the morning

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-3.48	-1.17	-3.62	-1.52	-3.73	-1.81	-3.85	-2.25	-4.14	-2.70	-5.53	-1.85	-5.66	-2.36	-5.76	-2.78	-5.89	-3.43	6.20	-4.03
Buy-Profits	0.15	1.20	0.05	0.55	0.13	1.27	0.14	1.62	0.14	2.80	0.16	1.26	0.07	0.66	0.14	1.39	0.15	1.70	0.11	2.20
Sell-Profits	-0.41	-4.43	-0.50	-5.13	-0.50	-5.33	-0.50	-5.83	0.04	0.90	-0.38	-3.96	-0.47	-4.67	-0.47	-4.85	-0.47	-5.30	0.06	1.46
Herfindhal	26.82	1.74	29.02	2.31	27.59	2.54	26.88	3.07	25.00	3.49	-6.63	-0.40	-4.39	-0.33	-5.77	-0.50	-6.33	-0.68	-8.14	-1.07
Dom/Ass	4.39	3.96	4.38	4.90	4.33	5.56	4.35	6.68	4.49	7.59	4.37	3.83	4.35	4.75	4.31	5.39	4.33	6.50	4.49	7.41
Volat Demand	-0.03	-1.29	-0.03	-1.75	-0.03	-1.85	-0.03	-2.21	-0.03	-2.35	0.007	0.25	0.003	0.17	0.006	0.35	0.007	0.46	0.009	0.70
Volat Assign	0.10	2.85	0.10	3.87	0.10	4.38	0.10	5.45	0.11	6.75	0.06	1.75	0.06	2.49	0.06	2.80	0.06	3.51	0.07	4.46
Volat Sec. Mkt	-0.48	-5.71	-0.49	-7.06	-0.48	-7.97	-0.47	-9.73	-0.49	-11.44	-0.36	-4.79	-0.37	-5.97	-0.36	-6.71	-0.36	-8.22	-0.37	-9.78
Duration	-0.31	-1.73	-0.32	-2.20	-0.32	-2.57	-0.32	-3.12	-0.40	-4.29	-0.05	-0.28	-0.05	-0.38	-0.05	-0.47	-0.06	-0.61	-0.13	-1.47
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.19		0.20		0.19		0.17		0.09		0.18		0.19		0.18		0.16		0.08	

Panel B: Profits from trades originated by the dealers in the afternoon

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-6.00	-2.12	-6.17	-2.64	-6.24	-3.06	-5.56	-3.26	-4.14	-2.70	-8.03	-2.84	-8.23	-3.55	-8.31	-4.11	-7.62	-4.47	-6.18	-4.02
Buy-Profits	2.88	2.65	3.17	2.92	3.56	3.05	3.13	3.28	0.39	0.69	2.94	2.69	3.25	2.96	3.64	3.09	3.18	3.31	0.39	0.70
Sell-Profits	-0.37	-0.72	-1.20	-1.40	-1.13	-1.38	-1.27	-1.61	-0.62	-1.85	-0.29	-0.58	-1.18	-1.36	-1.12	-1.34	-1.25	-1.56	-0.65	-2.18
Herfindhal	22.36	1.53	21.44	1.74	21.83	2.06	24.02	2.79	24.73	3.46	-10.77	-0.69	-11.72	-0.91	-11.31	-1.01	-9.06	-1.00	-8.32	-1.10
Dom/Ass	4.78	4.36	4.82	5.39	4.90	6.32	4.76	7.26	4.51	7.64	4.77	4.25	4.81	5.28	4.90	6.19	4.75	7.10	4.50	7.44
Volat Demand	-0.03	-1.11	-0.02	-1.23	-0.03	-1.56	-0.03	-1.97	-0.03	-2.34	0.01	0.44	0.01	0.61	0.01	0.58	0.01	0.64	0.009	0.70
Volat Assign	0.10	3.19	0.10	3.51	0.10	4.31	0.10	5.46	0.11	6.76	0.06	2.05	0.06	2.20	0.06	2.77	0.06	3.55	0.07	4.46
Volat Sec. Mkt	-0.39	-4.69	-0.38	-5.40	-0.38	-6.28	-0.42	-8.51	-0.49	-11.47	-0.28	-3.72	-0.26	-4.14	-0.26	-4.84	-0.30	-6.88	-0.37	-9.82
Duration	-0.37	-2.14	-0.36	-2.54	-0.38	-3.15	-0.38	-3.64	-0.40	-4.37	-0.11	-0.65	-0.09	-0.69	-0.12	-1.01	-0.11	-1.12	-0.14	-1.53
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.16		0.17		0.17		0.13		0.08		0.17		0.18		0.18		0.14		0.08	

Panel C: Profits from trades received by the dealers in the morning

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-4.43	-1.57	-4.23	-1.85	-4.29	-2.12	-4.19	-2.48	-4.13	-2.69	-6.46	-2.30	-6.26	-2.76	-6.32	-3.13	-6.22	-3.68	-6.18	-4.01
Buy-Profits	0.50	3.20	0.66	5.09	0.56	4.13	0.55	4.30	-0.03	-0.58	0.48	2.95	0.64	4.72	0.53	3.82	0.52	3.95	-0.06	-1.36
Sell-Profits	-0.12	-0.78	-0.04	-0.35	0.03	0.30	0.03	0.35	-0.21	-1.65	-0.13	-0.80	0.03	0.27	0.02	0.23	0.02	0.25	-0.26	-2.15
Herfindhal	23.98	1.53	24.12	1.90	26.70	2.44	27.47	3.11	24.70	3.45	-8.99	-0.54	-8.87	-0.66	-6.40	-0.55	-5.68	-0.61	-8.40	-1.11
Dom/Ass	4.58	4.37	4.59	5.46	4.49	5.89	4.47	6.98	4.51	7.62	4.56	4.27	4.57	5.34	4.47	5.74	4.46	6.79	4.49	7.43
Volat Demand	-0.03	-1.38	-0.03	-1.69	-0.03	-1.95	-0.03	-2.37	-0.03	-2.35	0.004	0.15	0.004	0.21	0.004	0.26	0.005	0.33	0.009	0.70
Volat Assign	0.10	3.01	0.10	3.76	0.10	4.54	0.11	5.68	0.11	6.76	0.06	1.97	0.06	2.46	0.06	3.00	0.07	3.76	0.07	4.47
Volat Sec. Mkt	-0.44	-5.33	-0.46	-6.71	-0.46	-7.79	-0.47	-9.73	-0.49	-11.45	-0.33	-4.43	-0.34	-5.62	-0.35	-6.55	-0.36	-8.25	-0.37	-9.80
Duration	-0.26	-1.52	-0.29	-2.13	-0.30	-2.38	-0.31	-3.00	-0.40	-4.35	-0.002	-0.01	-0.03	-0.22	-0.03	-0.29	-0.05	-0.50	-0.13	-1.51
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.23		0.25		0.20		0.17		0.08		0.23		0.25		0.20		0.17		0.08	

Panel D: Profits from trades received by the dealers in the afternoon

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-9.91	-3.84	-8.72	-3.98	-8.22	-4.21	-7.005	-4.26	-4.20	-2.74	-12.03	-4.73	-10.84	-5.03	-10.32	-5.35	-9.07	-5.55	-6.26	-4.07
Buy-Profits	2.83	2.41	1.73	1.75	1.60	1.88	1.96	2.48	4.05	1.42	2.91	2.47	1.75	1.75	1.61	1.87	1.94	2.42	4.50	1.43
Sell-Profits	-4.60	-4.86	-4.50	-4.95	-4.74	-5.36	-4.42	-5.53	2.43	2.25	-4.64	-4.82	-4.58	-4.96	-4.81	-5.37	-4.49	-5.55	3.10	2.66
Herfindhal	16.77	1.11	18.09	1.45	21.22	1.97	23.05	2.65	24.88	3.48	-16.38	-1.04	-15.08	-1.15	-11.89	-1.05	-10.05	-1.10	-8.11	-1.07
Dom/Ass	5.74	5.90	5.38	6.42	5.29	7.10	5.11	8.17	4.52	7.64	5.75	5.79	5.38	6.32	5.29	6.98	5.10	7.99	4.51	7.46
Volat Demand	-0.03	-1.36	-0.03	-1.33	-0.03	-1.70	-0.03	-2.18	-0.03	-2.37	0.005	0.20	0.01	0.50	0.008	0.42	0.007	0.44	0.009	0.65
Volat Assign	0.11	3.36	0.10	3.47	0.10	4.25	0.10	5.47	0.11	6.79	0.07	2.24	0.06	2.18	0.06	2.74	0.06	3.58	0.07	4.51
Volat Sec. Mkt	-0.23	-2.80	-0.26	-3.90	-0.29	-5.04	-0.35	-7.36	-0.49	-11.43	-0.10	-1.46	-0.13	-2.33	-0.17	-3.35	-0.24	-5.50	-0.37	-9.77
Herfindhal	-0.42	-2.70	-0.38	-2.94	-0.38	-3.24	-0.39	-3.99	-0.40	-4.36	-0.15	-1.02	-0.12	-0.94	-0.11	-1.01	-0.13	-1.34	-0.14	-1.54
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.29		0.25		0.22		0.19		0.08		0.30		0.26		0.23		0.19		0.08	

Panel E: Dealers' Net Profits in the morning

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-4.39	-1.44	-4.69	-1.93	-4.56	-2.15	-4.43	-2.58	-4.17	-2.73	-6.43	-2.10	-6.73	-2.75	-6.60	-3.09	-6.48	-3.68	-6.22	-4.05
Net Profits	-0.06	-3.67	-0.08	-4.10	-0.07	-4.05	-0.07	-4.20	0.02	2.64	-0.05	-3.33	-0.06	-3.80	-0.06	-3.74	-0.05	-3.91	0.01	2.26
Herfindhal	30.92	2.09	32.45	2.64	30.34	2.89	27.40	3.22	24.87	3.47	-2.68	-0.17	-1.01	-0.07	-2.98	-0.26	-5.74	-0.63	-8.20	-1.08
Dom/Ass	4.81	4.21	4.84	5.32	4.79	5.99	4.72	7.11	4.51	7.63	4.78	4.07	4.83	5.17	4.77	5.82	4.71	6.92	4.50	7.44
Volat Demand	-0.04	-1.76	-0.04	-2.27	-0.04	-2.38	-0.04	-2.56	-0.03	-2.33	-0.004	-0.15	-0.006	-0.30	-0.002	-0.13	0.001	0.12	0.00	0.70
Volat Assign	0.13	4.02	0.13	5.20	0.13	5.70	0.12	6.55	0.11	6.75	0.09	2.75	0.09	3.64	0.09	3.95	0.08	4.48	0.07	4.46
Volat Sec. Mkt	-0.52	-5.91	-0.52	-7.19	-0.51	-8.21	-0.50	-9.96	-0.49	-11.44	-0.40	-5.06	-0.40	-6.17	-0.39	-7.03	-0.39	-8.54	-0.37	-9.78
Duration	-0.42	-2.3	-0.42	-2.91	-0.42	-3.34	-0.42	-3.97	-0.40	-4.32	-0.15	-0.87	-0.15	-1.11	-0.16	-1.28	-0.15	-1.48	-0.13	-1.49
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.10		0.12		0.11		0.10		0.08		0.11		0.12		0.11		0.10		0.08	

Panel F: Dealers' Net Profits in the afternoon

Groups	(P _{t+1,A} - P _{t,B})										(P _{t+1,A} - P _{t,A})									
	G10		G15		G20		G30		Others		G10		G15		G2		G30		Others	
	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat	Value	t-stat
Constant	-4.23	-1.38	-3.97	-1.56	-4.09	-1.87	-4.10	-2.29	-4.15	-2.71	-6.28	-2.06	-6.00	-2.36	-6.12	-2.80	-6.14	-3.43	-6.20	-4.03
Net Profits	0.23	2.52	0.19	1.83	0.23	2.20	0.19	2.09	-0.08	-2.25	0.21	2.61	0.17	1.91	0.21	2.27	0.18	2.15	-0.07	-2.75
Herfindhal	22.73	1.55	22.52	1.88	22.46	2.16	23.61	2.80	24.84	3.47	-10.45	-0.68	-10.70	-0.85	-10.75	-0.98	-9.52	-1.07	-8.20	-1.08
Dom/Ass	4.57	3.95	4.49	4.65	4.53	5.45	4.51	6.60	4.51	7.64	4.56	3.87	4.47	4.54	4.52	5.32	4.50	6.43	4.50	7.44
Volat Demand	-0.03	-1.11	-0.02	-1.27	-0.02	-1.50	-0.03	-1.89	-0.03	-2.35	0.01	0.41	0.01	0.57	0.01	0.63	0.01	0.71	0.009	0.69
Volat Assign	0.11	3.33	0.11	3.90	0.11	4.57	0.11	5.66	0.11	6.77	0.07	2.20	0.07	2.53	0.07	2.99	0.07	3.71	0.07	4.48
Volat Sec. Mkt	-0.49	-5.54	-0.49	-6.87	-0.49	-7.87	-0.49	-9.73	-0.49	-11.46	-0.37	-4.71	-0.38	-5.87	-0.37	-6.71	-0.37	-8.31	-0.37	-9.81
Duration	-0.43	-2.43	-0.41	-2.74	-0.42	-3.25	-0.41	-3.86	-0.40	-4.37	-0.17	-0.98	-0.14	-0.99	-0.15	-1.23	-0.14	-1.40	-0.14	-1.53
Observat	1310		1965		2620		3930		5371		1310		1965		2620		3930		5371	
Arsquare	0.10		0.11		0.10		0.09		0.08		0.12		0.10		0.10		0.09		0.08	

TABLE 7 Market Depth

Market depth is calculated by estimating the equation $\Delta P_t = l q_t + y Z_t e_t$, where ΔP_t is the change in price at time t , q_t is the signed order flow at time t and Z_t denotes the sign of the incoming order at time t (+1 for a buyer-initiated trade and -1 for a seller-initiated trade). The error is assumed to be white noise. The specification is estimated for the *before* and *during* period separately for each auction and for the four days preceding each auction. We separately estimate market depth for the bond on auction and for all the other bonds not on auction. We report the means of the estimated l s across all the auctions for both the day of the auction (“Same Day”) and four the four preceding days (Auct.Day-1, ...). A *t-test* is performed to gauge the statistical significance of the difference among the auction and non-auction mean values for l . Given that the distributions of the resulting parameters (l) do not have the same variance, we use a *t-test* for unequal variance. $l_{auct,mor}^{auct}$ refers to the bond on auction, the day of the auction, in the morning. $l_{t,mor}^{auct}$ refers to the bond on auction, in the t th day before the auction, in the morning, $l_{auct,aft}^{auct}$ refers to the bond on auction, the day of the auction, in the afternoon. $l_{t,afternoon}^{auct}$ refers to the bond on auction, in the t th day before the auction, in the afternoon. Analogously $l_{auct,mor}^{nonauct}$, $l_{t,mor}^{nonauct}$, $l_{auct,aft}^{nonauct}$ $l_{t,afternoon}^{nonauct}$ refer to the bond non on auction.

Panel A

Days	$(l_{t,mor}^{auct} - l_{auct,mor}^{auct})$			$(l_{t,aft}^{auct} - l_{auct,aft}^{auct})$		
	Mean	t-stat	p-value	Mean	t-stat	p-value
Same Day	0.0010	-	-	0.0019	-	-
Auct.Day-1	0.0015	2.4885	0.0137	0.0011	-2.3848	0.0179
Auct.Day-2	0.0014	2.7953	0.0057	0.0011	-2.7216	0.0071
Auct.Day-3	0.0014	2.7200	0.0072	0.0009	-3.3737	0.0009
Auct.Day-4	0.0012	2.1020	0.0367	0.0011	-2.3493	0.0197

Days	$(l_{t,aft}^{nonauct} - l_{t,mor}^{nonauct})$			$(l_{t,aft}^{nonauct} - l_{t,mor}^{nonauct})$		
	Mean	t-stat	p-value	Mean	t-stat	p-value
Same Day	0.0011	3.8555	0.0002	-0.00022	-3.2843	0.0012
Auct.Day-1	0.0001	0.4090	0.6830	-0.00016	-2.6107	0.0096
Auct.Day-2	0.0004	1.7824	0.0762	-0.00028	-4.7993	0.0000
Auct.Day-3	0.0000	0.1428	0.8866	-0.00028	-4.8007	0.0000
Auct.Day-4	0.0006	1.9647	0.0511	-0.00036	-5.7344	0.0000

Panel B

$$\left(I_{t,mor}^{nonauct} - I_{t,aft}^{nonauct} \right) - \left(I_{auct,mor}^{auct} - I_{auct,aft}^{auct} \right) \quad \left(I_{t,mor}^{nonauct} - I_{t,aft}^{nonauct} \right) - \left(I_{t,mor}^{auct} - I_{t,aft}^{auct} \right)$$

Days	Mean	t-stat	p-value	Mean	t-stat	p-value
Same Day	0.00066	3.5186	0.0006	0.00066	3.5186	0.0006
Auct.Day-1	0.00072	3.7709	0.0002	-0.00015	-0.8073	0.4207
Auct.Day-2	0.00083	4.2294	0.00003	-0.00027	-1.3778	0.1702
Auct.Day-3	0.00088	4.4690	0.00001	0.00005	0.2934	0.7696
Auct.Day-4	0.00091	4.4621	0.00001	0.00019	1.1503	0.2516

$$\left(I_{t,mor}^{auct} - I_{t,aft}^{auct} \right) - \left(I_{auct,mor}^{auct} - I_{auct,aft}^{auct} \right) \quad \left(I_{t,mor}^{nonauct} - I_{t,aft}^{nonauct} \right) - \left(I_{auct,mor}^{noauct} - I_{auct,aft}^{noauct} \right)$$

Days	Mean	t-stat	p-value	Mean	t-stat	p-value
Same Day	-	-	-	-	-	-
Auct.Day-1	0.0009	3.3101	0.0011	-0.00001	-0.1848	0.8535
Auct.Day-2	0.0011	4.0496	0.0001	0.00005	0.6891	0.4913
Auct.Day-3	0.0008	3.3322	0.0010	0.00016	2.1003	0.0367
Auct.Day-4	0.0007	2.8760	0.0044	0.00014	1.8885	0.0600